

|| Project - Machine Learning ||

PROBLEM 1:

You are hired by one of the leading news channels CNBE who wants to analyze recent elections. This survey was conducted on 1525 voters with 9 variables. You have to build a model, to predict which party a voter will vote for on the basis of the given information, to create an exit poll that will help in predicting overall win and seats covered by a particular party.

DATA DICTIONARY:

1. vote: Party choice: Conservative or Labour
2. age: in years
3. economic.cond.national: Assessment of current national economic conditions, 1 to 5.
4. economic.cond.household: Assessment of current household economic conditions, 1 to 5.
5. Blair: Assessment of the Labour leader, 1 to 5.
6. Hague: Assessment of the Conservative leader, 1 to 5.
7. Europe: an 11-point scale that measures respondents' attitudes toward European integration. High scores represent 'Euroscptic' sentiment.
8. political.knowledge: Knowledge of parties' positions on European integration, 0 to 3.
9. gender: female or male.

1.1 Read the dataset. Do the descriptive statistics and do the null value condition check. Write an inference on it.

Importing the Required Libraries and Packages

```
In [1]: import numpy as np, pandas as pd, seaborn as sns, matplotlib.pyplot as plt
        %matplotlib inline
```

Loading the dataset and taking the first column as index

```
In [2]: df = pd.read_excel('D:\\SHUBHANK !\\GL\\7. TOPIC 6 - Machine Learning\\Final Project- M
```

Checking the Top 10 rows of the Dataset

```
In [3]: df.head(10)
```

```
Out[3]:
```

	vote	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	political.know
--	------	-----	------------------------	-------------------------	-------	-------	--------	----------------

	vote	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	political.knov
1	Labour	43	3	3	4	1	2	
2	Labour	36	4	4	4	4	5	
3	Labour	35	4	4	5	2	3	
4	Labour	24	4	2	2	1	4	
5	Labour	41	2	2	1	1	6	
6	Labour	47	3	4	4	4	4	
7	Labour	57	2	2	4	4	11	
8	Labour	77	3	4	4	1	1	
9	Labour	39	3	3	4	4	11	
10	Labour	70	3	2	5	1	11	

The dataset is loaded properly and the top 10 rows of the dataset are displayed.

As per the given data dictionary, we can also see that the columns 'economic.cond.national' and 'economic.cond.household' are having scales from 1 to 5, so here 1 means low and 5 means good.

Checking the 10 rows from the bottom

In [4]: `df.tail(10)`

Out[4]:

	vote	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	politi
1516	Conservative	82	2	2	2	1	11	
1517	Labour	30	3	4	4	2	4	
1518	Labour	76	4	3	2	2	11	
1519	Labour	50	3	4	4	2	5	
1520	Conservative	35	3	4	4	2	8	
1521	Conservative	67	5	3	2	4	11	
1522	Conservative	73	2	2	4	4	8	
1523	Labour	37	3	3	5	4	2	
1524	Conservative	61	3	3	1	4	11	
1525	Conservative	74	2	3	2	4	11	

The bottom 10 rows of the dataset are displayed.

Checking the shape(number of rows and columns) of the Dataset

In [5]: `print('Number of rows:',df.shape[0])`

```
print('Number of columns:',df.shape[1])
```

Number of rows: 1525

Number of columns: 9

Checking all the columns(variables) in the dataset

```
In [6]: df.columns
```

```
Out[6]: Index(['vote', 'age', 'economic.cond.national', 'economic.cond.household',
              'Blair', 'Hague', 'Europe', 'political.knowledge', 'gender'],
              dtype='object')
```

Checking the data types of all the variables present in the dataset

```
In [7]: df.dtypes
```

```
Out[7]: vote                object
age                int64
economic.cond.national  int64
economic.cond.household int64
Blair              int64
Hague              int64
Europe             int64
political.knowledge int64
gender             object
dtype: object
```

We can see that only 'vote' and 'gender' has the 'object' data types and all other variables are with 'integer' data types.

Describing the dataset

```
In [8]: df.describe()
```

```
Out[8]:
```

	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe
count	1525.000000	1525.000000	1525.000000	1525.000000	1525.000000	1525.000000
mean	54.182295	3.245902	3.140328	3.334426	2.746885	6.700000
std	15.711209	0.880969	0.929951	1.174824	1.230703	3.200000
min	24.000000	1.000000	1.000000	1.000000	1.000000	1.000000
25%	41.000000	3.000000	3.000000	2.000000	2.000000	4.000000
50%	53.000000	3.000000	3.000000	4.000000	2.000000	6.000000
75%	67.000000	4.000000	4.000000	4.000000	4.000000	10.000000
max	93.000000	5.000000	5.000000	5.000000	5.000000	11.000000

Few Observations based on above description/Summary :

1. We can see that maximum age of the voter is 93 and the minimum age is 24.
2. Mean age is 54.18 and Median age is 53.

- The standard deviation of the age is much higher than the standard deviation of the other features in the dataset.

Checking the Information about the dataset

In [9]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 1525 entries, 1 to 1525
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   vote                                1525 non-null   object
1   age                                1525 non-null   int64
2   economic.cond.national             1525 non-null   int64
3   economic.cond.household            1525 non-null   int64
4   Blair                              1525 non-null   int64
5   Hague                              1525 non-null   int64
6   Europe                             1525 non-null   int64
7   political.knowledge                1525 non-null   int64
8   gender                             1525 non-null   object
dtypes: int64(7), object(2)
memory usage: 119.1+ KB
```

- There are total 1525 rows and 9 columns.
- 7 variables are of integer data types and 2 are with object data types.
- There are no non-null values present in the dataset.

Checking for the duplicates

In [10]:

```
dups = df.duplicated()
print('Number of duplicate rows =', dups.sum())
df[dups]
```

Number of duplicate rows = 8

Out[10]:

	vote	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	politi
68	Labour	35	4	4	5	2	3	
627	Labour	39	3	4	4	2	5	
871	Labour	38	2	4	2	2	4	
984	Conservative	74	4	3	2	4	8	
1155	Conservative	53	3	4	2	2	6	
1237	Labour	36	3	3	2	2	6	
1245	Labour	29	4	4	4	2	2	
1439	Labour	40	4	3	4	2	2	

We found that 8 rows are duplicate in the dataset. Now we will remove them all.

Removing all the duplicates

```
In [11]: df.drop_duplicates(inplace=True)
```

Checking the shape of the dataset after removing the duplicates

```
In [12]: print('Number of rows:',df.shape[0])
         print('Number of columns:',df.shape[1])
```

```
Number of rows: 1517
Number of columns: 9
```

Checking the null values, if present

```
In [13]: df.isnull().sum()
```

```
Out[13]: vote                0
         age                0
         economic.cond.national  0
         economic.cond.household  0
         Blair              0
         Hague              0
         Europe             0
         political.knowledge  0
         gender             0
         dtype: int64
```

```
In [14]: # Drop the rows with 'nan' values

         df = df.dropna()

         df.shape
```

```
Out[14]: (1517, 9)
```

```
In [15]: df.shape
```

```
Out[15]: (1517, 9)
```

Let's check the location of presence of zeros in integer data type columns

```
In [16]: df.loc[((df['age'] == 0) | (df['economic.cond.national'] == 0) | (df['economic.cond.hou
              (df['Hague'] == 0) | (df['Europe'] == 0)))]
```

```
Out[16]:  vote  age  economic.cond.national  economic.cond.household  Blair  Hague  Europe  political.knowledge
```

```
In [17]: #Keeping the Original dataset safe, so copying

         original_data = df.copy()
```

1.2 Perform Univariate and Bivariate Analysis. Do exploratory data

analysis. Check for Outliers.

Let's check the value counts of 'vote' and 'gender' variable

```
In [18]: df['vote'].value_counts()
```

```
Out[18]: Labour          1057
Conservative      460
Name: vote, dtype: int64
```

```
In [19]: #percentage count

df.vote.value_counts(normalize=True)
```

```
Out[19]: Labour          0.69677
Conservative      0.30323
Name: vote, dtype: float64
```

We can see from the above value_counts that 'Labour' party has the more number of counts than 'Conservative' party.

```
In [20]: df['gender'].value_counts()
```

```
Out[20]: female      808
male          709
Name: gender, dtype: int64
```

```
In [21]: #percentage count

df.gender.value_counts(normalize=True)
```

```
Out[21]: female      0.53263
male          0.46737
Name: gender, dtype: float64
```

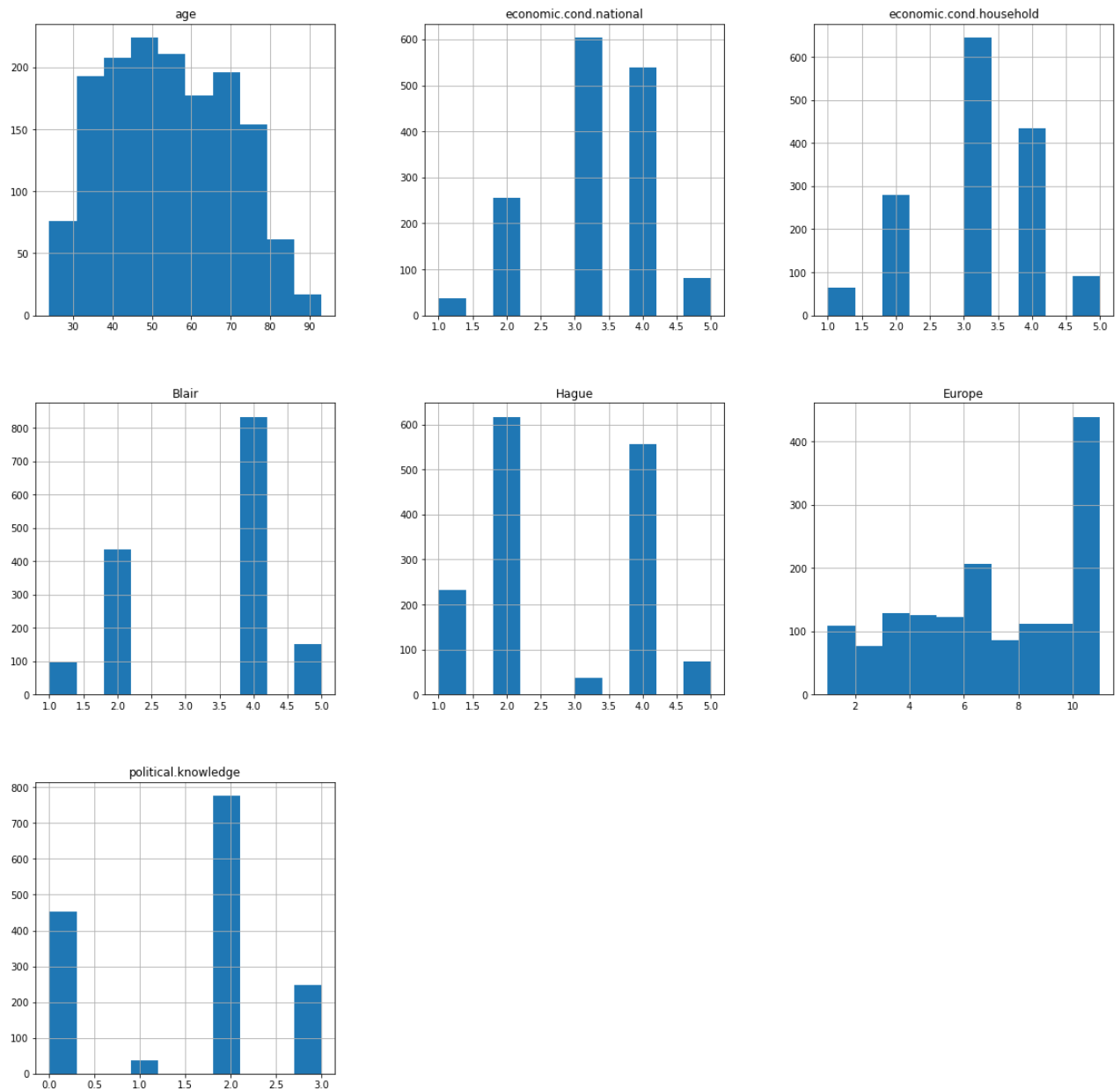
We can see that the female voters are more in count than the male voters.

Univariate and Bivariate Analysis

Histogram

```
In [22]: df.hist(figsize=(20,20))
```

```
Out[22]: array([[<AxesSubplot:title={'center':'age'}>,
      <AxesSubplot:title={'center':'economic.cond.national'}>,
      <AxesSubplot:title={'center':'economic.cond.household'}>],
      [<AxesSubplot:title={'center':'Blair'}>,
      <AxesSubplot:title={'center':'Hague'}>,
      <AxesSubplot:title={'center':'Europe'}>],
      [<AxesSubplot:title={'center':'political.knowledge'}>,
      <AxesSubplot:>, <AxesSubplot:>]], dtype=object)
```

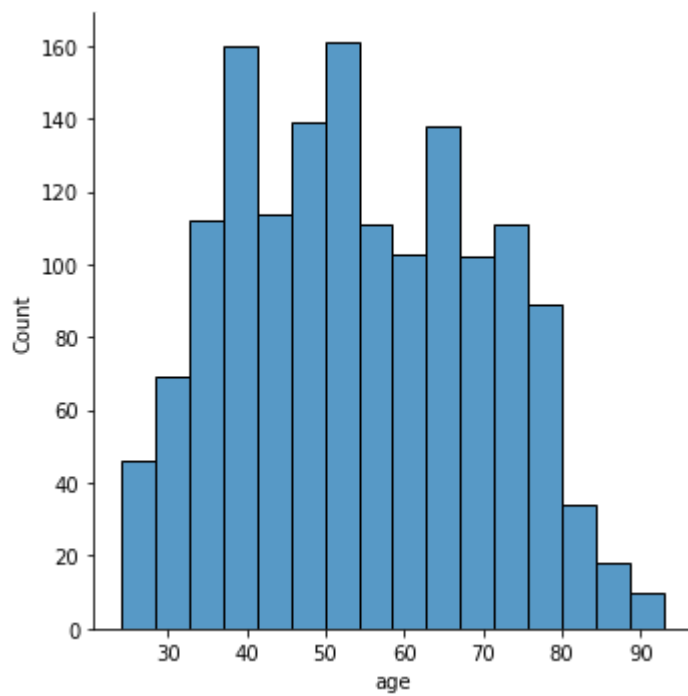


Distribution Plot to show the distribution of the continuous variables

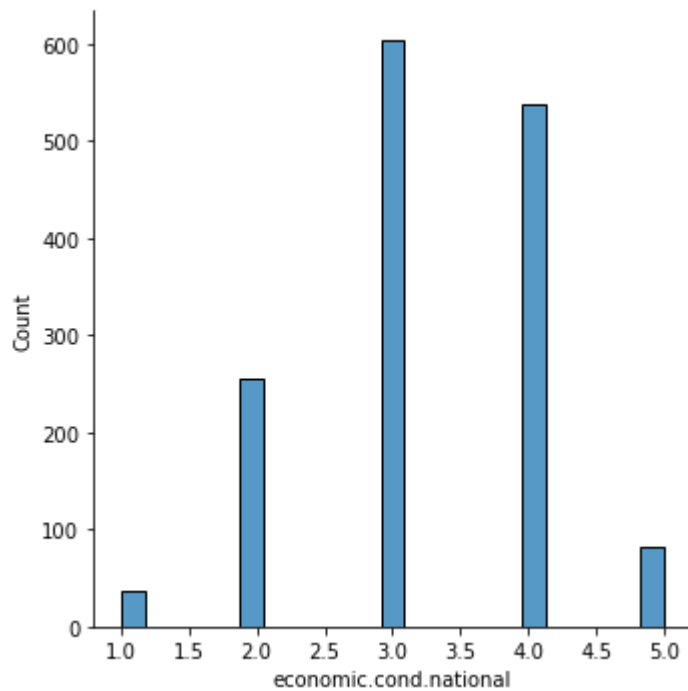
```
In [23]: num_columns = ['age', 'economic.cond.national', 'economic.cond.household',
                        'Blair', 'Hague', 'Europe', 'political.knowledge']

for i in num_columns:
    plt.figure()
    sns.displot(df[i])
    plt.show()
```

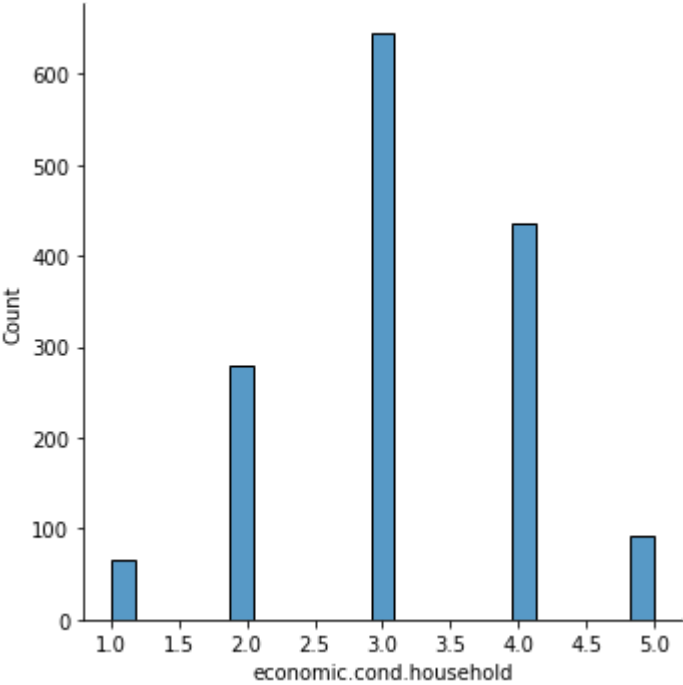
<Figure size 432x288 with 0 Axes>



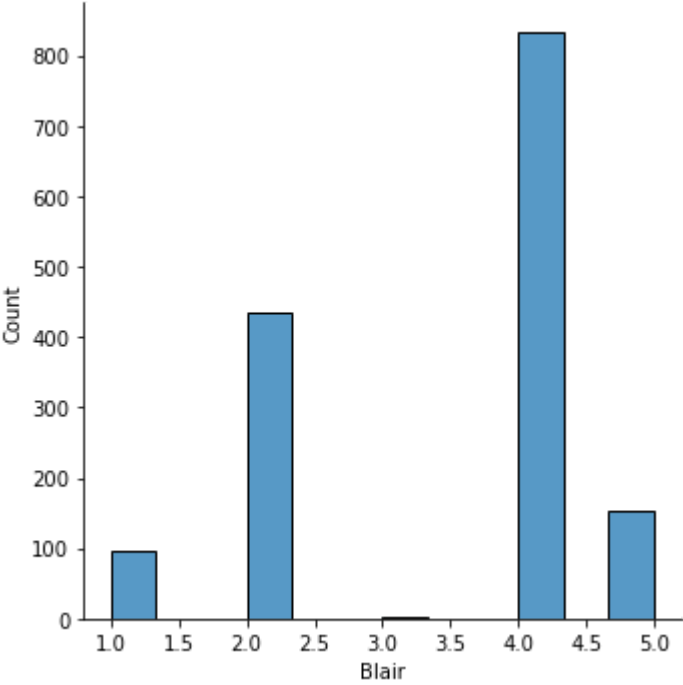
<Figure size 432x288 with 0 Axes>



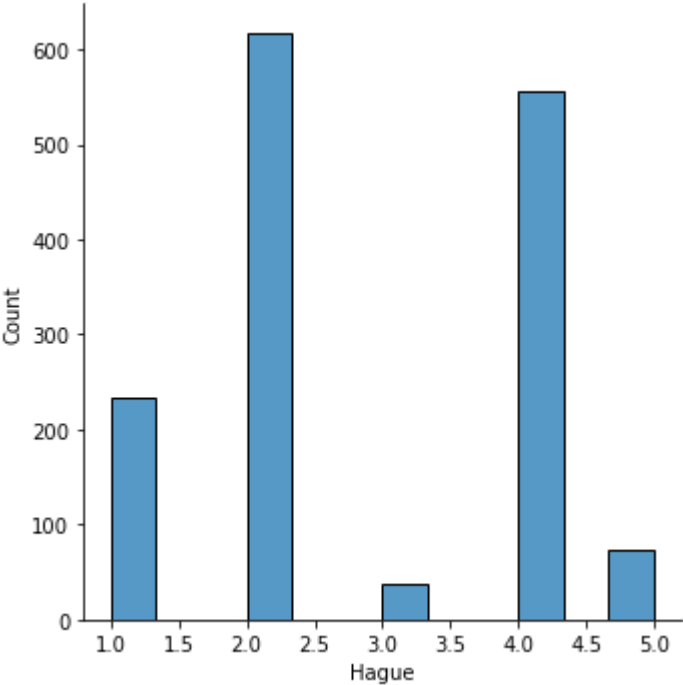
<Figure size 432x288 with 0 Axes>



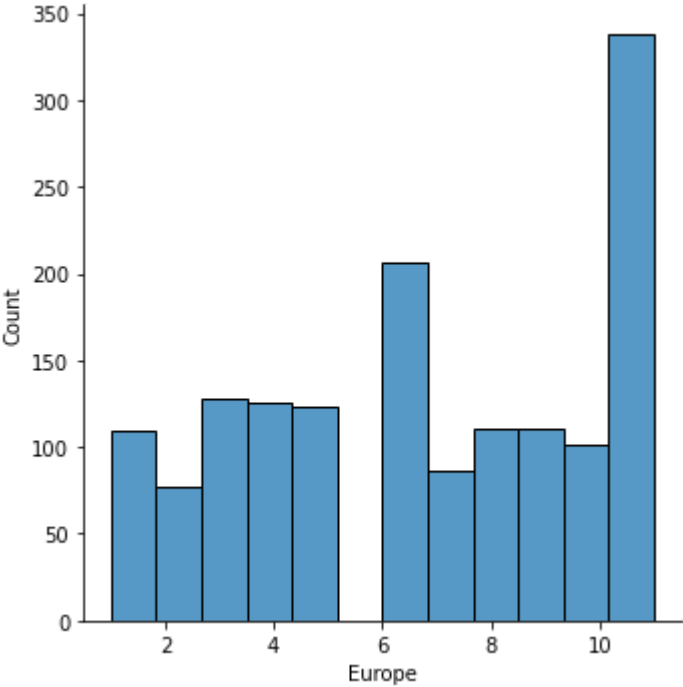
<Figure size 432x288 with 0 Axes>



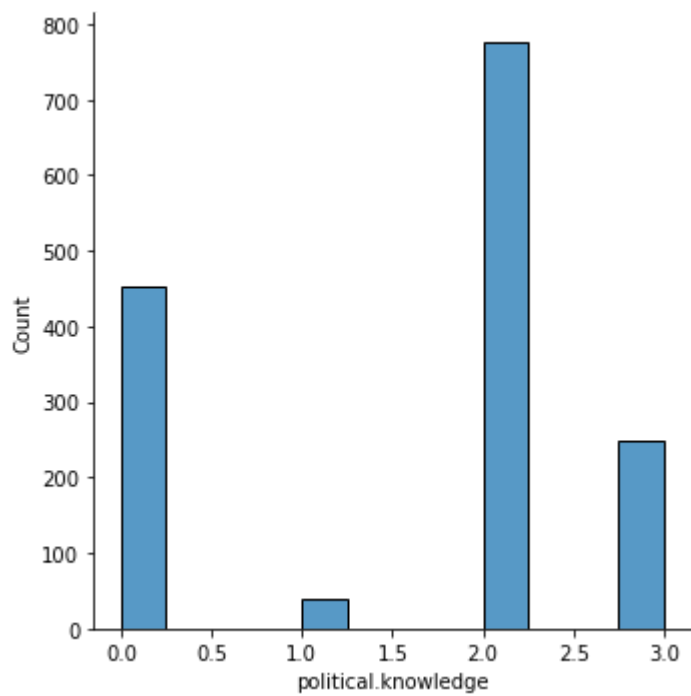
<Figure size 432x288 with 0 Axes>



<Figure size 432x288 with 0 Axes>

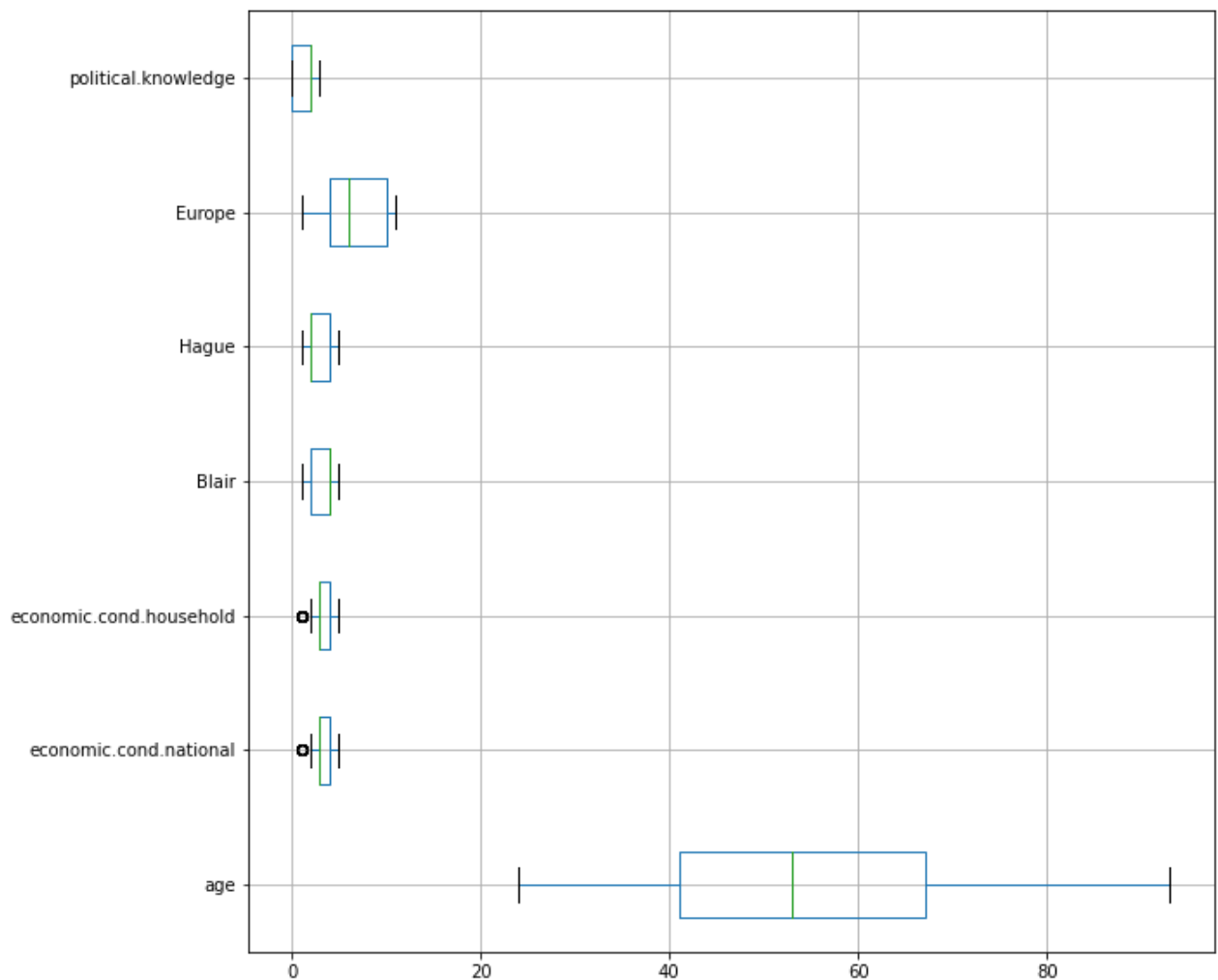


<Figure size 432x288 with 0 Axes>



Boxplot to show outliers

```
In [24]: plt.figure(figsize=(10,10))
df.boxplot(vert=0)
plt.show()
```

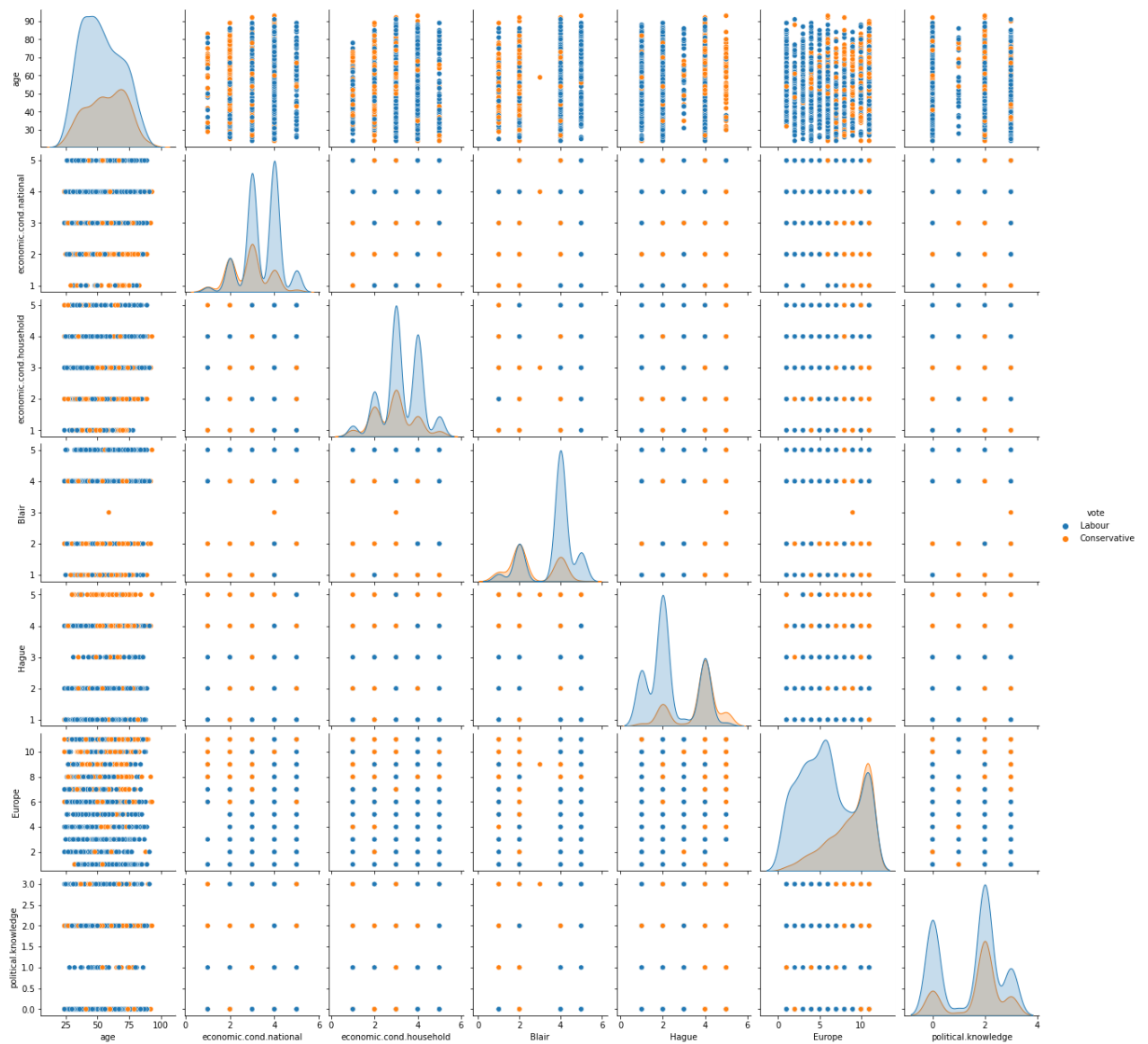


Plot is showing that very few Outliers are present in 'economic.cond.national' and 'economic.cond.household', and these are ratings or scale, so we can ignore and choose not to remove those outliers.

Pairplot

```
In [25]: sns.pairplot(df, hue='vote')
```

```
Out[25]: <seaborn.axisgrid.PairGrid at 0x13fe93d6b20>
```



Heatmap

```
In [26]: plt.figure(figsize=(12,9))
sns.heatmap(df.corr(),annot=True)
```

```
Out[26]: <AxesSubplot:>
```

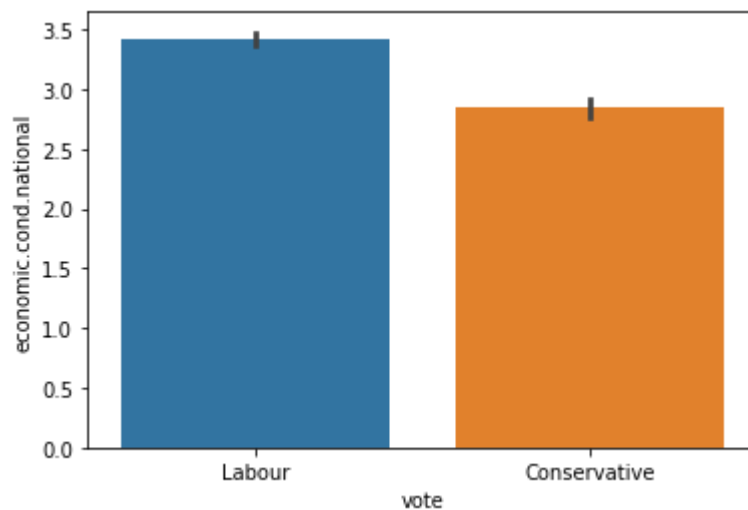


There isn't very high correlation between any two variables.

Barplot

```
In [27]: sns.barplot(data = df, x='vote',y='economic.cond.national')
```

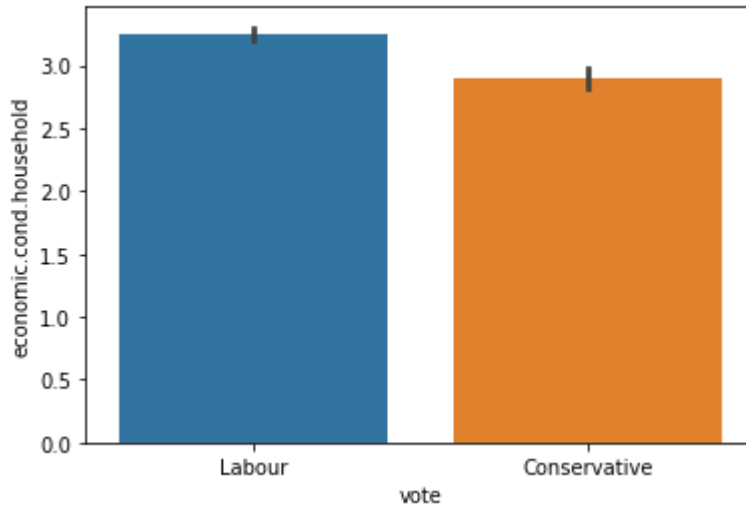
```
Out[27]: <AxesSubplot:xlabel='vote', ylabel='economic.cond.national'>
```



```
In [28]:
```

```
sns.barplot(data = df, x='vote',y='economic.cond.household')
```

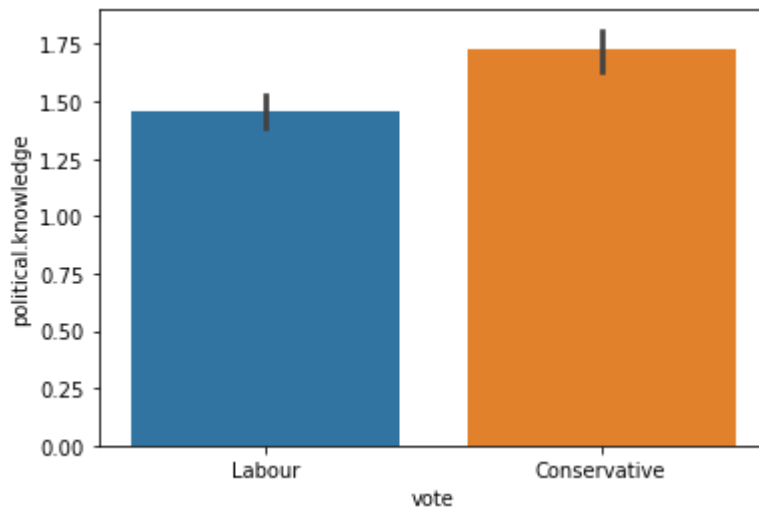
Out[28]: <AxesSubplot:xlabel='vote', ylabel='economic.cond.household'>



In [29]:

```
sns.barplot(data = df, x='vote',y='political.knowledge')
```

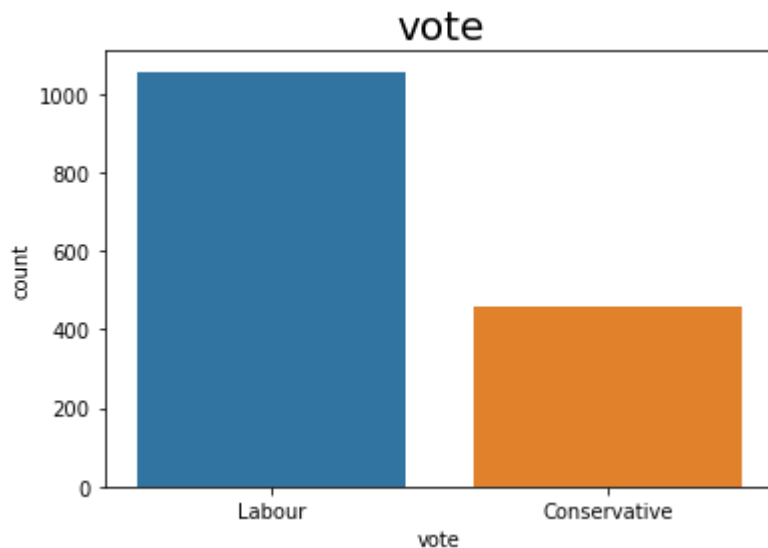
Out[29]: <AxesSubplot:xlabel='vote', ylabel='political.knowledge'>



Countplots

In [30]:

```
sns.countplot(data = df, x = df.vote)  
plt.title('vote', fontsize=20)  
plt.show()
```



We can see from the above countplot that Labour party gets more than 1000+ votes, which is much better than the vote counts of the Conservative party.

Checking categorical columns and numerical columns, so that the further calculations would be easy

```
In [31]: obj_cols=[]
int_cols=[]

for i in df.columns:
    if df[i].dtype=="object":
        obj_cols.append(i)
    else:
        int_cols.append(i)

print('Columns with Object datatype:\n',obj_cols)

print('Columns with Integer datatype:\n',int_cols)
```

Columns with Object datatype:

['vote', 'gender']

Columns with Integer datatype:

['age', 'economic.cond.national', 'economic.cond.household', 'Blair', 'Hague', 'Europe', 'political.knowledge']

We can see that python is treating categorical columns as numerical columns. But we know that except 'age' column, all other columns are categorical in nature. So, first we need to educate python that the columns like 'economic.cond.national', 'economic.cond.household', 'Blair', 'Hague', 'Europe', 'political.knowledge' and 'gender' are to be treated as categorical columns.

We will do this by type conversion.

```
In [32]: #using df.astype

df = df.astype({'vote':'category','economic.cond.national':'category', 'economic.cond.h

# Now print the data type of all columns after change
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 1517 entries, 1 to 1525
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   vote                                1517 non-null   category
1   age                                1517 non-null   int64
2   economic.cond.national             1517 non-null   category
3   economic.cond.household            1517 non-null   category
4   Blair                              1517 non-null   category
5   Hague                              1517 non-null   category
6   Europe                             1517 non-null   category
7   political.knowledge                 1517 non-null   category
8   gender                             1517 non-null   category
dtypes: category(8), int64(1)
memory usage: 69.5 KB
```

1.3 Encode the data (having string values) for Modeling. Is Scaling necessary here or not? Data Split: Split the data into train and test (70:30)

```
In [33]: cat_col = ['vote', 'gender']
```

```
In [34]: #Applying pd.get_dummies on cat_col

df = pd.get_dummies(df, columns=cat_col ,drop_first=True)
```

```
In [35]: df.head()
```

```
Out[35]:
```

	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	political.knowledge	v
1	43	3	3	4	1	2		2
2	36	4	4	4	4	5		2
3	35	4	4	5	2	3		2
4	24	4	2	2	1	4		0
5	41	2	2	1	1	6		2

```
In [36]: #keeping this df safe

df1 = df.copy()

dfcopy = df1.copy()
```

```
In [37]: #Now, Let's separate dependent and independent variables

X = df.drop('vote_Labour', axis=1)
Y = df[['vote_Labour']]
```

In [38]: `X.head()`

Out[38]:

	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	political.knowledge	g
1	43	3	3	4	1	2	2	
2	36	4	4	4	4	5	2	
3	35	4	4	5	2	3	2	
4	24	4	2	2	1	4	0	
5	41	2	2	1	1	6	2	

Scaling of the data

Scaling of the data comes under the set of steps of data pre-processing when we are performing machine learning algorithms in the data set. Data standardization is the process of rescaling the attributes so that they have mean as 0 and variance as 1. The ultimate goal to perform standardization is to bring down all the features to a common scale without distorting the differences in the range of the values.

To bring all features in the same standing, we need to do scaling so that one significant number does not impact the model just because of their large magnitude.

Features scaling is necessary for this dataset, because we have seen in the boxplot that the 'age' feature has the numeric values which are much greater than the values present in the other features of the dataset.

Also we found out from the description(describe function) of the dataset that standard deviation of the 'age' is much higher than the standard deviation of the other available features in the dataset.

So, to normalize the range of independent variables or features of data, we will be scaling our data before any further analysis.

In [39]:

```
#Using standard scaler:

from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

scaler.fit(X) #we will use scaled_df for the models that gets affected without feature
```

Out[39]: StandardScaler()

In [40]: `X.head()`

Out[40]:

	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	political.knowledge	g
1	43	3	3	4	1	2	2	
2	36	4	4	4	4	5	2	

	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	political.knowledge	g
3	35	4		4	5	2	3	2
4	24	4		2	2	1	4	0
5	41	2		2	1	1	6	2

Split the data into train and test set in 70:30 ratio

```
In [41]: from sklearn.model_selection import train_test_split

X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.3 , random_state=
```

```
In [42]: X_train.head()
```

```
Out[42]:
```

	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	political.knowledge	
992	34	2		4	1	4	11	2
1275	40	4		3	4	4	6	0
650	61	4		3	4	4	7	2
678	47	3		3	4	2	11	0
539	44	5		3	4	2	8	0

```
In [43]: scaler.fit_transform(X_test)
X_test.head()
```

```
Out[43]:
```

	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	political.knowledge	
505	71	3		3	2	2	8	2
370	43	3		2	4	2	8	3
1076	89	5		5	5	2	1	2
1032	47	2		3	2	4	8	2
1330	33	5		4	4	4	8	0

1.4. Apply Logistic Regression and LDA (linear discriminant analysis).

Logistic Regression

```
In [44]: #We will use GridSearchCV for getting the best parameters for the model
```

```
from sklearn.model_selection import GridSearchCV

grid={'penalty':['l1','l2','none'], 'solver':['lbfgs', 'liblinear'], 'tol':[0.0001,0.00
```

```
In [45]: # Fit the Logistic Regression model

from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=100000,n_jobs=2)

grid_search = GridSearchCV(estimator = model, param_grid = grid, cv = 3,n_jobs=-1,scori
```

```
In [46]: grid_search.fit(X_train, Y_train)
```

C:\Users\user\anaconda3\lib\site-packages\sklearn\model_selection_search.py:922: UserWarning: One or more of the test scores are non-finite: [nan nan 0.88492887 0.88478031 0.8842122 0.8842122 0.88508645 0.88508645 0.8842122 0.8842122 nan nan]
warnings.warn(

C:\Users\user\anaconda3\lib\site-packages\sklearn\utils\validation.py:63: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
return f(*args, **kwargs)

C:\Users\user\anaconda3\lib\site-packages\sklearn\linear_model_logistic.py:1355: UserWarning: 'n_jobs' > 1 does not have any effect when 'solver' is set to 'liblinear'. Got 'n_jobs' = 2.
warnings.warn("'n_jobs' > 1 does not have any effect when"

```
Out[46]: GridSearchCV(cv=3, estimator=LogisticRegression(max_iter=100000, n_jobs=2),
              n_jobs=-1,
              param_grid={'penalty': ['l1', 'l2', 'none'],
                           'solver': ['lbfgs', 'liblinear'],
                           'tol': [0.0001, 1e-06]},
              scoring='f1')
```

```
In [47]: #Getting the best parameters and best grid estimators:

grid_search.best_params_
```

```
Out[47]: {'penalty': 'l2', 'solver': 'liblinear', 'tol': 0.0001}
```

```
In [48]: best_model = grid_search.best_estimator_
best_model
```

```
Out[48]: LogisticRegression(max_iter=100000, n_jobs=2, solver='liblinear')
```

Predicting on Training and Test dataset

```
In [49]: ytrain_predict = best_model.predict(X_train)
ytest_predict = best_model.predict(X_test)
```

```
In [50]: ## Getting the probabilities on the test set
```

```
ytest_predict_prob = best_model.predict_proba(X_test)
```

```
In [51]: #Checking the head from the predicted probability
pd.DataFrame(ytest_predict_prob).head(10)
```

```
Out[51]:
```

	0	1
0	0.427675	0.572325
1	0.155830	0.844170
2	0.006244	0.993756
3	0.842250	0.157750
4	0.066400	0.933600
5	0.062857	0.937143
6	0.349056	0.650944
7	0.230190	0.769810
8	0.043898	0.956102
9	0.137002	0.862998

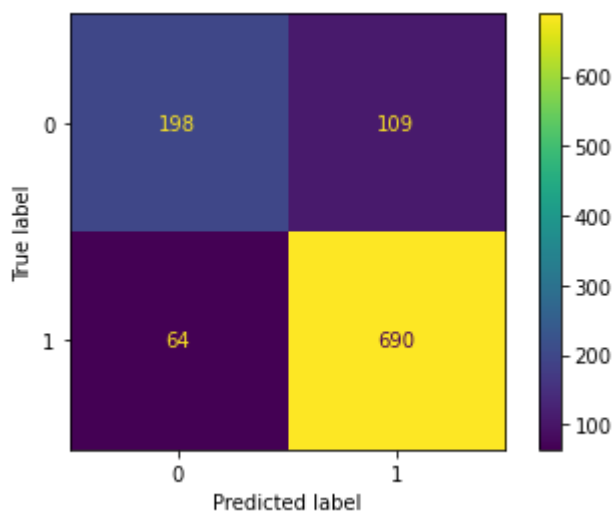
Confusion Matrix and Classification Report on the Training Data

```
In [52]: ## Confusion matrix on the training data

from sklearn import metrics
from sklearn.metrics import roc_auc_score,roc_curve,classification_report,confusion_mat

plot_confusion_matrix(best_model,X_train,Y_train)
```

```
Out[52]: <sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x13fecfa7a30>
```



```
In [53]: print(classification_report(Y_train, ytrain_predict))
```

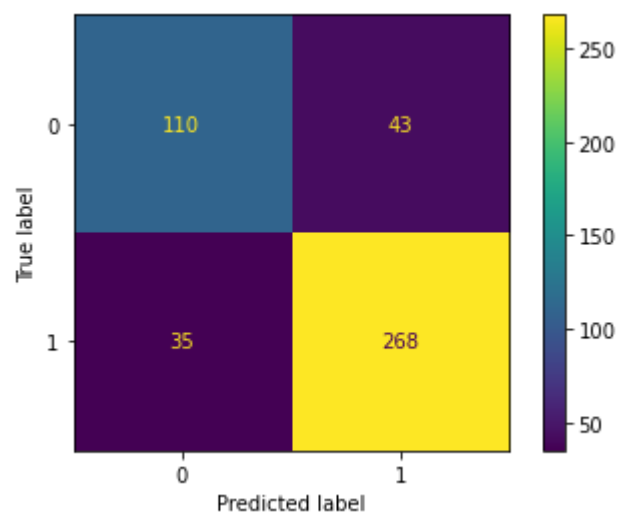
	precision	recall	f1-score	support
0	0.76	0.64	0.70	307
1	0.86	0.92	0.89	754
accuracy			0.84	1061
macro avg	0.81	0.78	0.79	1061
weighted avg	0.83	0.84	0.83	1061

Confusion Matrix and Classification Report on Testing Data

```
In [54]: ## Confusion matrix on the test data

plot_confusion_matrix(best_model,X_test,Y_test)
```

```
Out[54]: <sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x13fed262130>
```



```
In [55]: print(classification_report(Y_test, ytest_predict))
```

	precision	recall	f1-score	support
0	0.76	0.72	0.74	153
1	0.86	0.88	0.87	303
accuracy			0.83	456
macro avg	0.81	0.80	0.81	456
weighted avg	0.83	0.83	0.83	456

Conclusion

The accuracy of the model on train data is low than that of test data, that means the model is good to go.

The recall and f1 score also good in case of training data.

Model Evaluation

```
In [56]: # Accuracy score for training data

acc_train = best_model.score(X_train, Y_train)

acc_train
```

Out[56]: 0.8369462770970783

AUC and ROC for the training data

```
In [57]: # predict probabilities

probs = best_model.predict_proba(X_train)
```

```
In [58]: # keep probabilities for the positive outcome only

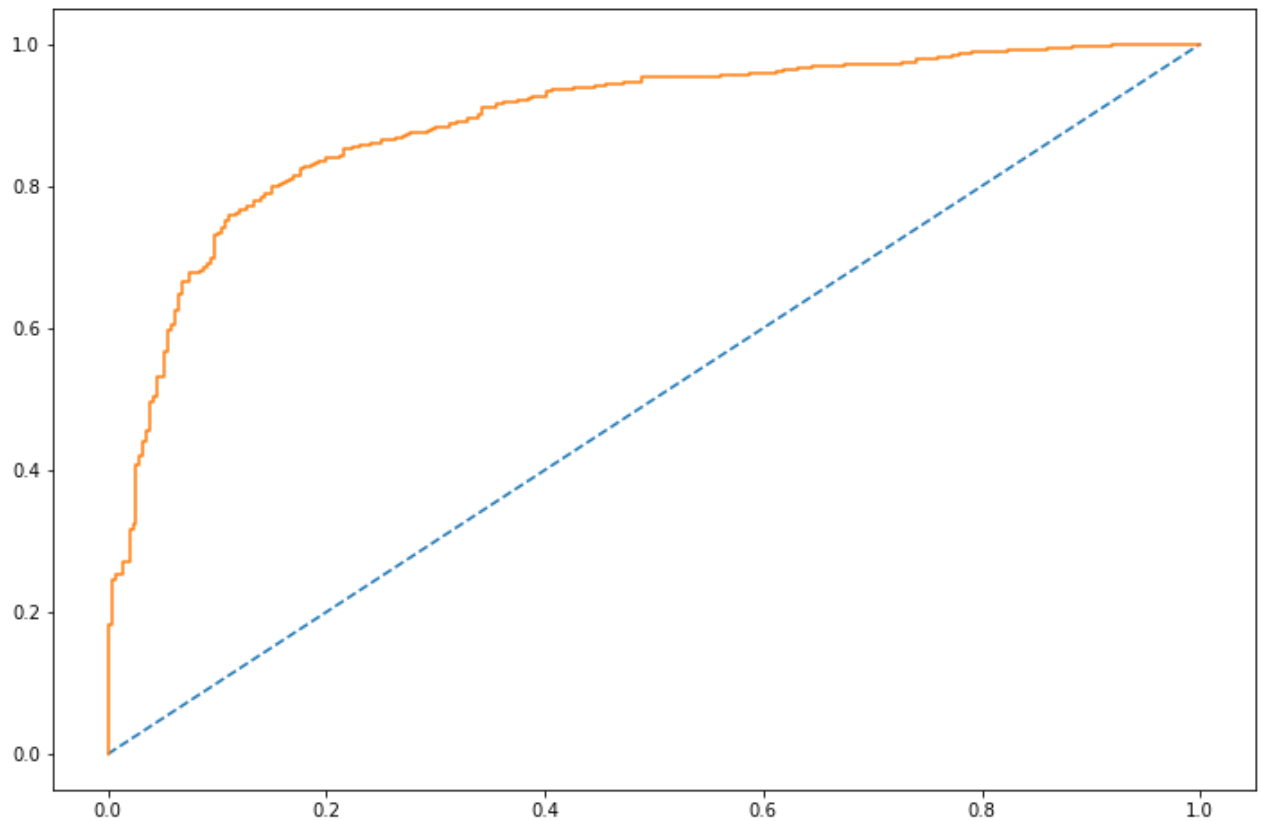
probs = probs[:, 1]
```

```
In [59]: # calculate AUC
auc = roc_auc_score(Y_train, probs)
print('AUC: %.3f' % auc)
```

AUC: 0.890

```
In [60]: # calculate roc curve

train_fpr, train_tpr, train_thresholds = roc_curve(Y_train, probs)
plt.rcParams["figure.figsize"] = (12,8)
plt.plot([0, 1], [0, 1], linestyle='--')
# plot the roc curve for the model
plt.plot(train_fpr, train_tpr);
```



```
In [61]: #Accuracy score for test data

acc_test = best_model.score(X_test, Y_test)

acc_test
```

Out[61]: 0.8289473684210527

```
In [62]: # predict probabilities
probs = best_model.predict_proba(X_test)

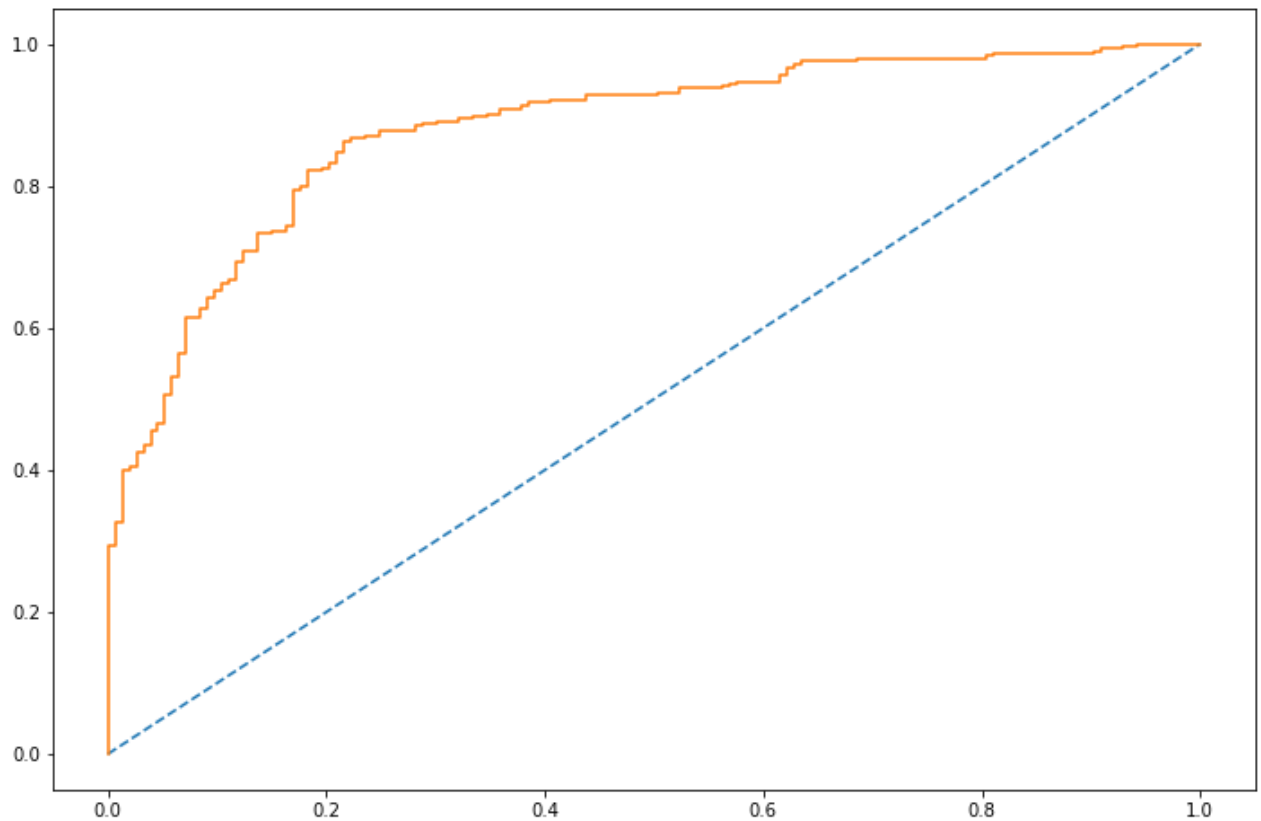
# Probabilities for the positive outcome only
probs = probs[:, 1]

# calculate AUC
test_auc = roc_auc_score(Y_test, probs)
print('AUC: %.3f' % test_auc)

# calculate roc curve
test_fpr, test_tpr, test_thresholds = roc_curve(Y_test, probs)
plt.plot([0, 1], [0, 1], linestyle='--')

# plot the roc curve for the model
plt.plot(test_fpr, test_tpr);
```

AUC: 0.880



Linear Discriminant Analysis (LDA)

```
In [63]: from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
from sklearn import metrics, model_selection
from sklearn.preprocessing import scale

x = df1.drop('vote_Labour',axis=1)
y = df1.pop('vote_Labour')
```

```
In [64]: from sklearn.model_selection import train_test_split,GridSearchCV

x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.20,random_state=1,stra
```

```
In [65]: clf = LinearDiscriminantAnalysis()

LDA_model=clf.fit(x_train,y_train)
```

```
In [66]: #Train data class prediction
#cut off value is 0.5

pred_class_train = LDA_model.predict(x_train)
```

```
In [67]: # Test data class prediction
#cut-off value is 0.5

pred_class_test = LDA_model.predict(x_test)
```

```
In [68]: print('Classification Report of the training data:\n\n',metrics.classification_report(y_train,pred_prob_train))
print('Classification Report of the test data:\n\n',metrics.classification_report(y_test,pred_prob_test))
```

Classification Report of the training data:

	precision	recall	f1-score	support
0	0.74	0.68	0.71	368
1	0.86	0.90	0.88	845
accuracy			0.83	1213
macro avg	0.80	0.79	0.79	1213
weighted avg	0.83	0.83	0.83	1213

Classification Report of the test data:

	precision	recall	f1-score	support
0	0.80	0.72	0.75	92
1	0.88	0.92	0.90	212
accuracy			0.86	304
macro avg	0.84	0.82	0.83	304
weighted avg	0.86	0.86	0.86	304

Probability prediction for the training and test data

```
In [69]: # Training Data Probability Prediction
pred_prob_train = LDA_model.predict_proba(x_train)

# Test Data Probability Prediction
pred_prob_test = LDA_model.predict_proba(x_test)
```

AUC and ROC for the training data

```
In [70]: # calculate AUC
auc = metrics.roc_auc_score(y_train,pred_prob_train[:,1])
print('AUC for the Training Data: %.3f' % auc)

# calculate roc curve
fpr, tpr, thresholds = metrics.roc_curve(y_train,pred_prob_train[:,1])
plt.plot([0, 1], [0, 1], linestyle='--')
# plot the roc curve for the model
plt.plot(fpr, tpr, marker='.',label = 'Training Data')

# AUC and ROC for the test data

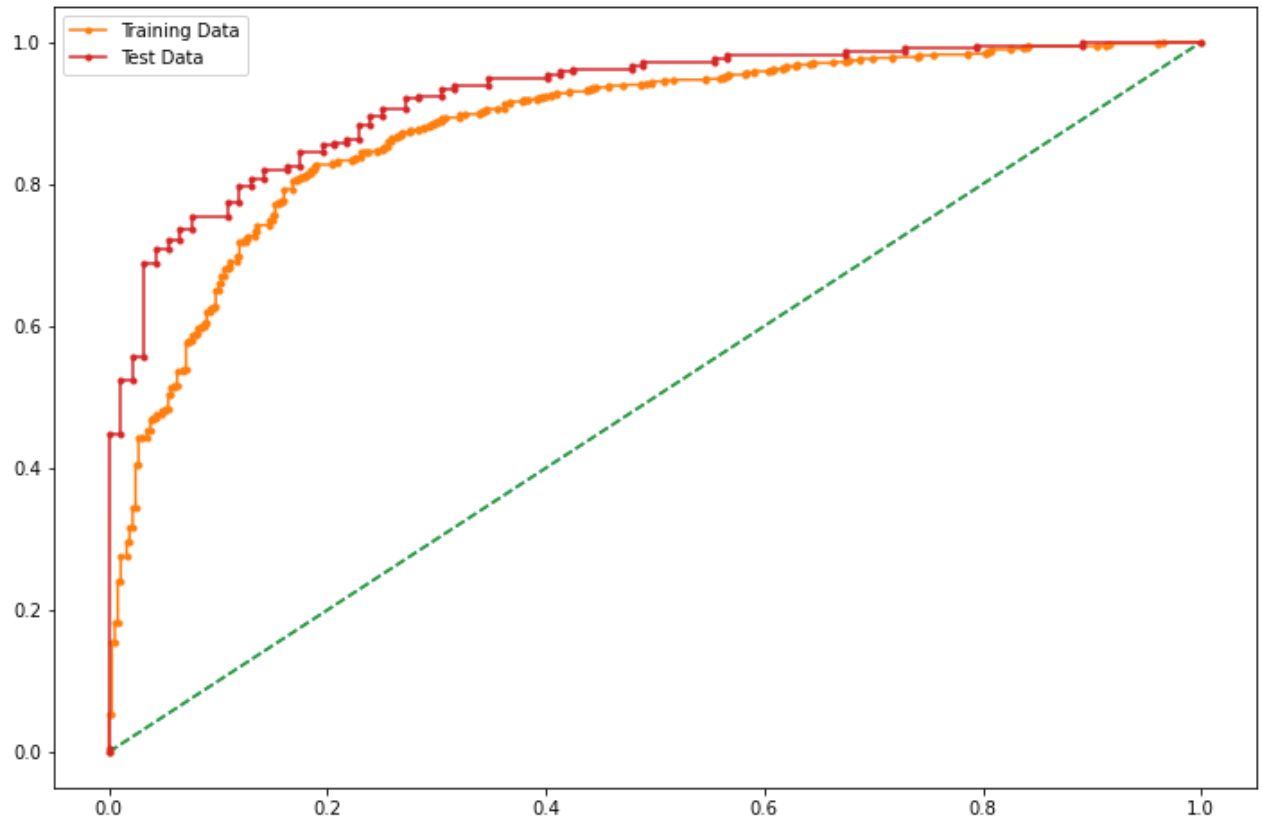
# calculate AUC
auc = metrics.roc_auc_score(y_test,pred_prob_test[:,1])
print('AUC for the Test Data: %.3f' % auc)

# calculate roc curve
fpr, tpr, thresholds = metrics.roc_curve(y_test,pred_prob_test[:,1])
```

```
plt.plot([0, 1], [0, 1], linestyle='--')
# plot the roc curve for the model
plt.plot(fpr, tpr, marker='.', label='Test Data')
# show the plot
plt.legend(loc='best')
plt.show()
```

AUC for the Training Data: 0.880

AUC for the Test Data: 0.921



Conclusion

The accuracy of the model for the train data is 83% and for the testing data is 86%.

recall and f1 score is also pretty good in testing data.

1.5 Apply KNN Model and Naïve Bayes Model. Interpret the results.

KNN Model

```
In [71]: from scipy.stats import zscore
```

```
In [72]: X[['age', 'economic.cond.national', 'economic.cond.household', 'Blair', 'Hague', 'Europe', 'p
```

```
In [73]: X.head()
```

```
Out[73]:
```

	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	politica
--	-----	------------------------	-------------------------	-------	-------	--------	----------

	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	politica
1	-0.716161	-0.278185	-0.148020	0.565802	-1.419969	-1.437338	
2	-1.162118	0.856242	0.926367	0.565802	1.014951	-0.527684	
3	-1.225827	0.856242	0.926367	1.417312	-0.608329	-1.134120	
4	-1.926617	0.856242	-1.222408	-1.137217	-1.419969	-0.830902	
5	-0.843577	-1.412613	-1.222408	-1.988727	-1.419969	-0.224465	

```
In [74]: #Now we will split the data into train and test set for KNN model

from sklearn.model_selection import train_test_split

X_train,X_test,Y_train,Y_test=train_test_split(X,Y, random_state=1)
```

```
In [75]: #Now Lets import KNeighboursClassifier

from sklearn.neighbors import KNeighborsClassifier

KNN_model = KNeighborsClassifier()

KNN_model.fit(X_train,Y_train)
```

C:\Users\user\anaconda3\lib\site-packages\sklearn\neighbors_classification.py:179: Data ConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().

```
return self._fit(X, y)
Out[75]: KNeighborsClassifier()
```

```
In [76]: #Let's predict the model on X_train and check the model score

predicted_labels = KNN_model.predict(X_train)

KNN_model_score = KNN_model.score(X_train,Y_train)
```

```
In [77]: #model score

print(KNN_model_score)
```

0.8566402814423922

Confusion mAtrix and Classification Report

```
In [78]: print('Confusion Matrix\n')

print(metrics.confusion_matrix(Y_train,predicted_labels))

print('Classification Report\n')

print(metrics.classification_report(Y_train,predicted_labels))
```

Confusion Matrix

```
[[232  95]
 [ 68 742]]
```

Classification Report

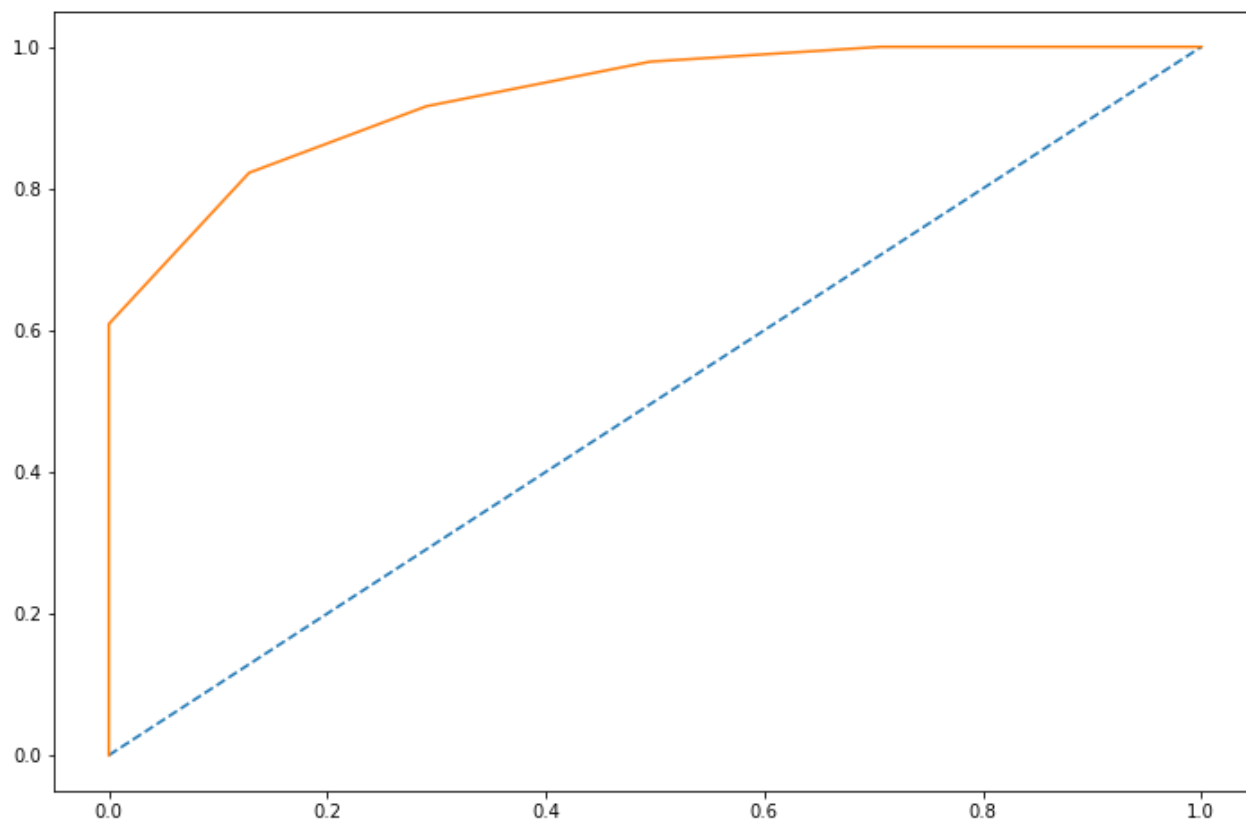
	precision	recall	f1-score	support
0	0.77	0.71	0.74	327
1	0.89	0.92	0.90	810
accuracy			0.86	1137
macro avg	0.83	0.81	0.82	1137
weighted avg	0.85	0.86	0.85	1137

In [79]:

```
### AUC ROC Curve on Train data
```

```
probs_train=KNN_model.predict_proba(X_train)
probs_train=probs_train[:, 1]
auc=roc_auc_score(Y_train,probs_train)
print("auc %.3f " % auc)
train_fpr ,train_tpr, train_thresholds= roc_curve(Y_train,probs_train)
plt.plot([0,1],[0,1],linestyle='--')
plt.plot(train_fpr,train_tpr);
```

auc 0.929



Naive Bayes Model

In [80]:

```
df.head()
```

```
Out[80]:
```

	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	political.knowledge	v
1	43	3	3	4	1	2		2
2	36	4	4	4	4	5		2
3	35	4	4	5	2	3		2
4	24	4	2	2	1	4		0
5	41	2	2	1	1	6		2

```
In [81]: #Let's first separate the independent and dependent variables from the saved dataset tha
X = df.drop('vote_Labour',axis=1)
Y = df['vote_Labour']
x_train, x_test, y_train, y_test = train_test_split(X, Y, test_size=0.3, random_state=1)
x_train.head()
```

```
Out[81]:
```

	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	political.knowledge	
992	34	2	4	1	4	11		2
1275	40	4	3	4	4	6		0
650	61	4	3	4	4	7		2
678	47	3	3	4	2	11		0
539	44	5	3	4	2	8		0

Train Naive Bayes algorithm

```
In [82]: from sklearn.naive_bayes import GaussianNB # using Gaussian algorithm from Naive Bayes
# create the model
nb_model = GaussianNB()
nb_model.fit(x_train, y_train.ravel())
```

```
Out[82]: GaussianNB()
```

Let see the performance of our model with training data

```
In [83]: nb_train_predict = nb_model.predict(x_train)
from sklearn import metrics
print("Model Accuracy: {:.2f}".format(metrics.accuracy_score(y_train, nb_train_predict))
```

Model Accuracy: 0.84

Now let see the performance of our model with testing data

```
In [84]: nb_test_predict = nb_model.predict(x_test)

from sklearn import metrics

print("Model Accuracy: {:.2f}".format(metrics.accuracy_score(y_test, nb_test_predict)))
```

Model Accuracy: 0.82

Confusion Matrix and Classification Report

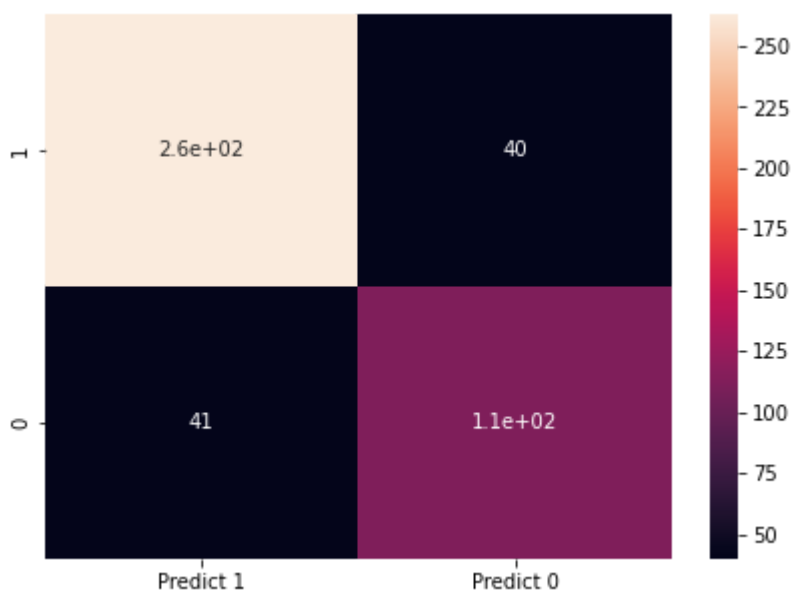
```
In [85]: print("Confusion Matrix:\n")

cm = metrics.confusion_matrix(y_test, nb_test_predict, labels=[1, 0])

df_cm = pd.DataFrame(cm, index = [i for i in ["1","0"]],
                      columns = [i for i in ["Predict 1","Predict 0"]])
plt.figure(figsize = (7,5))
sns.heatmap(df_cm, annot=True)
```

Confusion Matrix:

Out[85]: <AxesSubplot:>



```
In [86]: print("Classification Report")
print(metrics.classification_report(y_test, nb_test_predict, labels=[1, 0]))
```

```
Classification Report
              precision    recall  f1-score   support

     1       0.87       0.87       0.87        303
     0       0.74       0.73       0.73        153

 accuracy                   0.82        456
```

macro avg	0.80	0.80	0.80	456
weighted avg	0.82	0.82	0.82	456

The precision for 1 is above 80% and accuracy is also above 80%

1.6 Model Tuning, Bagging (Random Forest should be applied for Bagging), and Boosting.

```
In [87]: #We will use Random forest for Bagging
#Let's first import RandomForestClassifier from sklearn.ensemble

from sklearn.ensemble import RandomForestClassifier
```

```
In [88]: #we will take the copied df that is original_data for building Random Forest
#because as we know that Random Forest is not affected by feature scaling

dfcopy.head()
```

```
Out[88]:
```

	age	economic.cond.national	economic.cond.household	Blair	Hague	Europe	political.knowledge	v
1	43	3	3	4	1	2		2
2	36	4	4	4	4	5		2
3	35	4	4	5	2	3		2
4	24	4	2	2	1	4		0
5	41	2	2	1	1	6		2

```
In [89]: dfcopy.info()

<class 'pandas.core.frame.DataFrame'>
Int64Index: 1517 entries, 1 to 1525
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   age                                    1517 non-null   int64
1   economic.cond.national                1517 non-null   category
2   economic.cond.household               1517 non-null   category
3   Blair                                 1517 non-null   category
4   Hague                                 1517 non-null   category
5   Europe                                1517 non-null   category
6   political.knowledge                   1517 non-null   category
7   vote_Labour                           1517 non-null   uint8
8   gender_male                           1517 non-null   uint8
dtypes: category(6), int64(1), uint8(2)
memory usage: 69.2 KB
```

```
In [90]: dfcopy = dfcopy.astype({'economic.cond.national': 'category', 'economic.cond.household':
```

```
In [91]: dfcopy.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 1517 entries, 1 to 1525
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   age                                    1517 non-null   int64
1   economic.cond.national                1517 non-null   category
2   economic.cond.household               1517 non-null   category
3   Blair                                 1517 non-null   category
4   Hague                                 1517 non-null   category
5   Europe                                1517 non-null   category
6   political.knowledge                   1517 non-null   category
7   vote_Labour                           1517 non-null   category
8   gender_male                           1517 non-null   category
dtypes: category(8), int64(1)
memory usage: 69.5 KB
```

```
In [92]: X = dfcopy.drop("vote_Labour" , axis=1)
        y = dfcopy.pop("vote_Labour")
```

```
In [93]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=.30, random_state=1)
```

Now Let's Build RandomForest

```
In [94]: rfcl = RandomForestClassifier(n_estimators=301, oob_score = True, max_depth = 10, max_f
        rfcl = rfcl.fit(X_train, y_train)
```

NOTE:

n_estimators is the number of trees, that we want to build within the RandomForestClassifier. Out of bag (OOB) score is a way of validating the Random forest model. OOB score is computed as the number of correctly predicted rows from the out of bag sample. max_features will be used to select randomly features from the available number of features.

```
In [95]: rfcl
```

```
Out[95]: RandomForestClassifier(max_depth=10, max_features=5, min_samples_leaf=50,
                               min_samples_split=110, n_estimators=301, oob_score=True)
```

```
In [96]: rfcl.oob_score_
```

```
Out[96]: 0.823751178133836
```

```
In [97]: #Checking the error rate in %:

        print('error_rate(in %) :', 1- (rfcl.oob_score_))

        #NOTE: OOB Error is the number of wrongly classifying the OOB Sample.

        error_rate(in %) : 0.17624882186616397
```

For the above specified grid parameters, we got the rfcl model with oob score 80.6% and error rate 19.3%, which shows the model is good to go for further evaluation.

Getting the Feature Importances

```
In [98]: print (pd.DataFrame(rfcl.feature_importances_, columns = ["Imp"],
                        index = X_train.columns).sort_values('Imp',ascending=False))
```

	Imp
Hague	0.406048
Blair	0.302594
Europe	0.205316
economic.cond.national	0.038313
political.knowledge	0.021143
age	0.019404
economic.cond.household	0.005826
gender_male	0.001356

Let's Now use Ensemble Learning methods and see the result

We will use Bagging , Ada Boosting and GradientBoost

Let's first look by use Bagging:

```
In [99]: #First import BaggingClassifier from sklearn.ensemble

#we will be using random forest as base_estimator

from sklearn.ensemble import BaggingClassifier

bgcl = BaggingClassifier(base_estimator=rfcl, n_estimators=50,random_state=1)
#bgcl = BaggingClassifier(n_estimators=50,random_state=1)

bgcl = bgcl.fit(X_train, y_train)
```

```
In [100... y_predict = bgcl.predict(X_test)

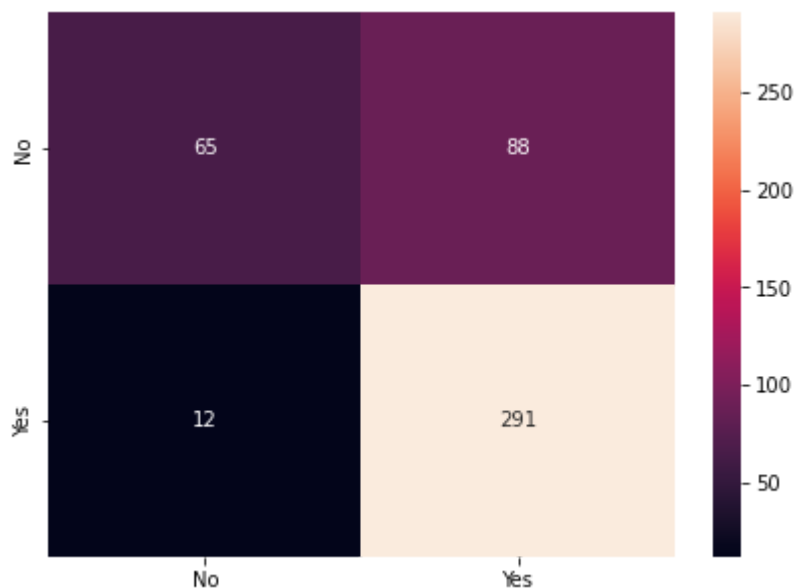
print(bgcl.score(X_test , y_test))

cm=metrics.confusion_matrix(y_test, y_predict,labels=[0, 1])

df_cm = pd.DataFrame(cm, index = [i for i in ["No","Yes"]],
                    columns = [i for i in ["No","Yes"]])
plt.figure(figsize = (7,5))
sns.heatmap(df_cm, annot=True ,fmt='g')
```

0.7807017543859649

Out[100... <AxesSubplot:>



Ada Boosting

```
In [101... #first importing AdaBoostClassifier from sklearn.ensemble
from sklearn.ensemble import AdaBoostClassifier

abcl = AdaBoostClassifier(n_estimators=10, random_state=1)

#Applying .fit to the train data
abcl = abcl.fit(X_train, y_train)
```

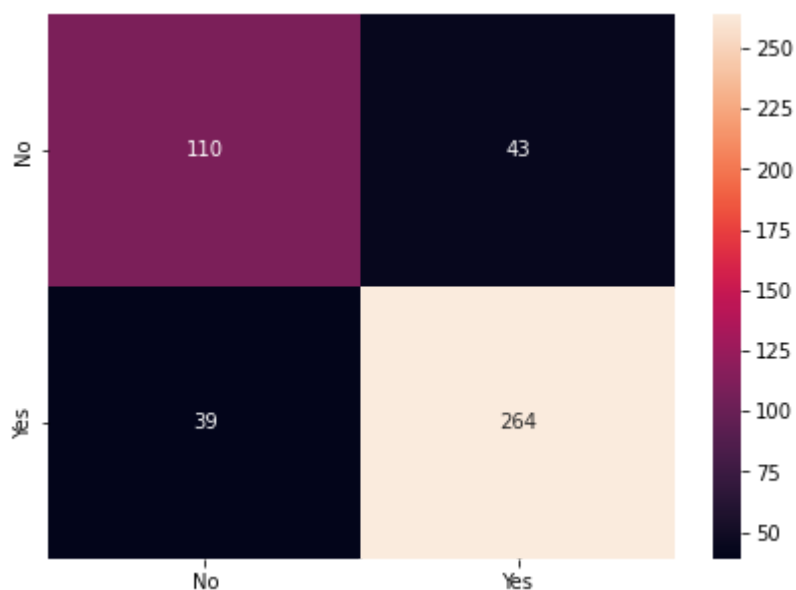
```
In [102... y_predict = abcl.predict(X_test)
print(abcl.score(X_test , y_test))

cm=metrics.confusion_matrix(y_test, y_predict, labels=[0, 1])

df_cm = pd.DataFrame(cm, index = [i for i in ["No", "Yes"]],
                      columns = [i for i in ["No", "Yes"]])
plt.figure(figsize = (7,5))
sns.heatmap(df_cm, annot=True ,fmt='g')
```

0.8201754385964912

Out[102... <AxesSubplot:>



GradientBoost

```
In [103... from sklearn.ensemble import GradientBoostingClassifier
gbcl = GradientBoostingClassifier(n_estimators = 50, random_state=1)
gbcl = gbcl.fit(X_train, y_train)
```

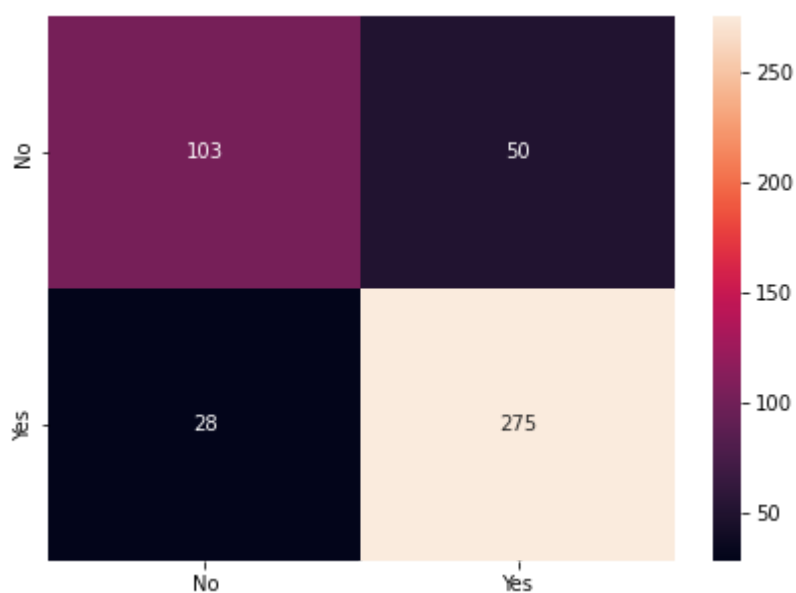
```
In [104... y_predict = gbcl.predict(X_test)
print(gbcl.score(X_test, y_test))
cm=metrics.confusion_matrix(y_test, y_predict, labels=[0, 1])

df_cm = pd.DataFrame(cm, index = [i for i in ["No", "Yes"]],
                      columns = [i for i in ["No", "Yes"]])
plt.figure(figsize = (7,5))
sns.heatmap(df_cm, annot=True, fmt='g')
```

0.8289473684210527

<AxesSubplot:>

Out[104...



1.7 Performance Metrics: Check the performance of Predictions on Train and Test sets using Accuracy, Confusion Matrix, Plot ROC curve and get ROC_AUC score for each model. Final Model: Compare the models and write inference which model is best/optimized. (7 marks)

Performance Metrics has already been performed on models

1. Logistic Regression: For Logistic Regression on train data, we can see the below Classification Report:

The intention behind using logistic regression is to find the best fitting model to describe the relationship between the dependent and the independent variable.

The accuracy and recall value on the train data is above 80% which is a good indication for the Labour Party.

	precision	recall	f1-score	support
0	0.76	0.64	0.70	307
1	0.86	0.92	0.89	754
accuracy			0.84	1061

macro avg 0.81 0.78 0.79 1061 weighted avg 0.83 0.84 0.83 1061

Same model when fit on test data:

	precision	recall	f1-score	support
0	0.76	0.72	0.74	153
1	0.86	0.88	0.87	303
accuracy			0.83	456

macro avg 0.81 0.80 0.81 456 weighted avg 0.83 0.83 0.83 456

For train data, Area under curve (AUC) = 89%

And for test data , AUC: 88%

1. For Linear Discriminant Analysis

In Linear Discriminant Analysis (LDA):

Classification Report of the training data:

	precision	recall	f1-score	support
0	0.74	0.68	0.71	368
1	0.86	0.90	0.88	845
accuracy			0.83	1213

macro avg 0.80 0.79 0.79 1213 weighted avg 0.83 0.83 0.83 1213

Classification Report of the test data:

	precision	recall	f1-score	support
0	0.80	0.72	0.75	92
1	0.88	0.92	0.90	212
accuracy			0.86	304

macro avg 0.84 0.82 0.83 304 weighted avg 0.86 0.86 0.86 304

AUC for the Training Data: 88% AUC for the Test Data: 92%

1. For KNN model:

The model score is 85%

Classification Report:

	precision	recall	f1-score	support
0	0.77	0.71	0.74	327
1	0.89	0.92	0.90	810
accuracy			0.86	1137

macro avg 0.83 0.81 0.82 1137 weighted avg 0.85 0.86 0.85 1137

AUC on train data = auc 0.929

1. Naive Bayes:

In train data model accuracy Model Accuracy: 0.84

On test data Model Accuracy: 0.82

Classification Report:

	precision	recall	f1-score	support
1	0.87	0.87	0.87	303
0	0.74	0.73	0.73	153

accuracy

0.82

456

macro avg 0.80 0.80 0.80 456 weighted avg 0.82 0.82 0.82 456

The precision for 1 is above 80% and accuracy is also above 80%.

1.8 Based on these predictions, what are the insights?

Insights:

By looking at the insights from the above predictions, we can confidently say that Labour Party would be the first choice to win in the elections. Below are some insights :

1. We can see that the Labour party will get more number of votes than the conservative party.
2. Blair, who is the leader from Labour party has received good ratings than the Hague from conservative party.
3. On a scale of 1 to 5, Blair has got rating 4 more than Hague in the assessment.
4. Female candidates are more active in voting than male.
5. Age does not have much impact on the result.
6. In all the models and their prediction on test data, the classification report seems to be in favour of 1(Labour Party votes)
7. On the basis of calculations based on current economic household condition and national economic condition, the Labour party has got better performance representation than its opponent party.

-----THE END -----

Problem 2:

In this particular project, we are going to work on the inaugural corpora from the nltk in Python. We will be looking at the following speeches of the Presidents of the United States of America:

1. President Franklin D. Roosevelt in 1941
2. President John F. Kennedy in 1961
3. President Richard Nixon in 1973

Loading the data

```
In [105...  ## Importing the necessary Libraries along with the standard import

import numpy as np
import pandas as pd

#Importing the Natural Language Tool Kit(nltk) which contains a lot of functionalities
import nltk
import random

# this is used for string manipulations
import string
```

```
#To download 'stopwords' and 'punctuations(punkt)'
#nltk.download('stopwords')
#nltk.download('punkt')

#inaugural is already downloaded
#nltk.download('inaugural')

from nltk.corpus import inaugural
inaugural.fileids()

Roosevelt_raw = inaugural.raw('1941-Roosevelt.txt')
Kennedy_raw = inaugural.raw('1961-Kennedy.txt')
Nixon_raw = inaugural.raw('1973-Nixon.txt')
```

2.1 Find the number of characters, words, and sentences for the mentioned documents.

In [106... *#First let's check all the raw files separately*

Roosevelt_raw

Out[106... 'On each national day of inauguration since 1789, the people have renewed their sense of dedication to the United States.\n\nIn Washington\'s day the task of the people was to create and weld together a nation.\n\nIn Lincoln\'s day the task of the people was to preserve that Nation from disruption from within.\n\nIn this day the task of the people is to save that Nation and its institutions from disruption from without.\n\nTo us there has come a time, in the midst of swift happenings, to pause for a moment and take stock -- to recall what our place in history has been, and to rediscover what we are and what we may be. If we do not, we risk the real peril of inaction.\n\nLives of nations are determined not by the count of years, but by the lifetime of the human spirit. The life of a man is three-score years and ten: a little more, a little less. The life of a nation is the fullness of the measure of its will to live.\n\nThere are men who doubt this. There are men who believe that democracy, as a form of Government and a frame of life, is limited or measured by a kind of mystical and artificial fate that, for some unexplained reason, tyranny and slavery have become the surging wave of the future -- and that freedom is an ebbing tide.\n\nBut we Americans know that this is not true.\n\nEight years ago, when the life of this Republic seemed frozen by a fatalistic terror, we proved that this is not true. We were in the midst of shock -- but we acted. We acted quickly, boldly, decisively.\n\nThese later years have been living years -- fruitful years for the people of this democracy. For they have brought to us greater security and, I hope, a better understanding that life\'s ideals are to be measured in other than material things.\n\nMost vital to our present and our future is this experience of a democracy which successfully survived crisis at home; put away many evil things; built new structures on enduring lines; and, through it all, maintained the fact of its democracy.\n\nFor action has been taken within the three-way framework of the Constitution of the United States. The coordinate branches of the Government continue freely to function. The Bill of Rights remains inviolate. The freedom of elections is wholly maintained. Prophets of the downfall of American democracy have seen their dire predictions come to naught.\n\nDemocracy is not dying.\n\nWe know it because we have seen it revive--and grow.\n\nWe know it cannot die -- because it is built on the unhampered initiative of individual men and women joined together in a common enterprise -- an enterprise undertaken and carried through by the free expression of a free majority.\n\nWe know it because democracy alone, of all forms of government, enlists the full force of men\'s enlightened will.\n\nWe know it because democracy alone has constructed an unlimited civilization capable of infinite progress in the improvement of human life.\n\nWe know it because, if we look below the surface, we sense it still spreading on every continent -- for it is the most humane, the most advanced, and in the end the most unconquerable of all forms of human society.\n\nA nation, like a

person, has a body--a body that must be fed and clothed and housed, invigorated and rested, in a manner that measures up to the objectives of our time.

A nation, like a person, has a mind -- a mind that must be kept informed and alert, that must know itself, that understands the hopes and the needs of its neighbors -- all the other nations that live within the narrowing circle of the world.

And a nation, like a person, has something deeper, something more permanent, something larger than the sum of all its parts. It is that something which matters most to its future -- which calls forth the most sacred guarding of its present.

It is a thing for which we find it difficult -- even impossible -- to hit upon a single, simple word.

And yet we all understand what it is -- the spirit -- the faith of America. It is the product of centuries. It was born in the multitudes of those who came from many lands -- some of high degree, but mostly plain people, who sought here, early and late, to find freedom more freely.

The democratic aspiration is no mere recent phase in human history. It is human history. It permeated the ancient life of early peoples. It blazed anew in the middle ages. It was written in Magna Charta.

In the Americas its impact has been irresistible. America has been the New World in all tongues, to all peoples, not because this continent was a new-found land, but because all those who came here believed they could create upon this continent a new life -- a life that should be new in freedom.

Its vitality was written into our own Mayflower Compact, into the Declaration of Independence, into the Constitution of the United States, into the Gettysburg Address.

Those who first came here to carry out the longings of their spirit, and the millions who followed, and the stock that sprang from them -- all have moved forward constantly and consistently toward an ideal which in itself has gained stature and clarity with each generation.

The hopes of the Republic cannot forever tolerate either undeserved poverty or self-serving wealth.

We know that we still have far to go; that we must more greatly build the security and the opportunity and the knowledge of every citizen, in the measure justified by the resources and the capacity of the land.

But it is not enough to achieve these purposes alone. It is not enough to clothe and feed the body of this Nation, and instruct and inform its mind. For there is also the spirit. And of the three, the greatest is the spirit.

Without the body and the mind, as all men know, the Nation could not live.

But if the spirit of America were killed, even though the Nation's body and mind, constricted in an alien world, lived on, the America we know would have perished.

That spirit -- that faith -- speaks to us in our daily lives in ways often unnoticed, because they seem so obvious. It speaks to us here in the Capital of the Nation. It speaks to us through the processes of governing in the sovereignties of 48 States. It speaks to us in our counties, in our cities, in our towns, and in our villages. It speaks to us from the other nations of the hemisphere, and from those across the seas -- the enslaved, as well as the free. Sometimes we fail to hear or heed these voices of freedom because to us the privilege of our freedom is such an old, old story.

The destiny of America was proclaimed in words of prophecy spoken by our first President in his first inaugural in 1789 -- words almost directed, it would seem, to this year of 1941: "The preservation of the sacred fire of liberty and the destiny of the republican model of government are justly considered deeply, finally, staked on the experiment intrusted to the hands of the American people."

If we lose that sacred fire--if we let it be smothered with doubt and fear -- then we shall reject the destiny which Washington strove so valiantly and so triumphantly to establish. The preservation of the spirit and faith of the Nation does, and will, furnish the highest justification for every sacrifice that we may make in the cause of national defense.

In the face of great perils never before encountered, our strong purpose is to protect and to perpetuate the integrity of democracy.

For this we muster the spirit of America, and the faith of America.

We do not retreat. We are not content to stand still. As Americans, we go forward, in the service of our country, by the will of God.

In [107...

#Raw file of Kennedy

Kennedy_raw

Out[107...

'Vice President Johnson, Mr. Speaker, Mr. Chief Justice, President Eisenhower, Vice President Nixon, President Truman, reverend clergy, fellow citizens, we observe today not a victory of party, but a celebration of freedom -- symbolizing an end, as well as a beginning -- signifying renewal, as well as change. For I have sworn I before you and Almighty

y God the same solemn oath our forebears I prescribed nearly a century and three quarters ago.

The world is very different now. For man holds in his mortal hands the power to abolish all forms of human poverty and all forms of human life. And yet the same revolutionary beliefs for which our forebears fought are still at issue around the globe -- the belief that the rights of man come not from the generosity of the state, but from the hand of God.

We dare not forget today that we are the heirs of that first revolution. Let the word go forth from this time and place, to friend and foe alike, that the torch has been passed to a new generation of Americans -- born in this century, tempered by war, disciplined by a hard and bitter peace, proud of our ancient heritage -- and unwilling to witness or permit the slow undoing of those human rights to which this Nation has always been committed, and to which we are committed today at home and around the world.

Let every nation know, whether it wishes us well or ill, that we shall pay any price, bear any burden, meet any hardship, support any friend, oppose any foe, in order to assure the survival and the success of liberty.

This much we pledge -- and more.

To those old allies whose cultural and spiritual origins we share, we pledge the loyalty of faithful friends. United, there is little we cannot do in a host of cooperative ventures. Divided, there is little we can do -- for we dare not meet a powerful challenge at odds and split asunder.

To those new States whom we welcome to the ranks of the free, we pledge our word that one form of colonial control shall not have passed away merely to be replaced by a far more iron tyranny. We shall not always expect to find them supporting our view. But we shall always hope to find them strongly supporting their own freedom -- and to remember that, in the past, those who foolishly sought power by riding the back of the tiger ended up inside.

To those peoples in the huts and villages across the globe struggling to break the bonds of mass misery, we pledge our best efforts to help them help themselves, for whatever period is required -- not because the Communists may be doing it, not because we seek their votes, but because it is right. If a free society cannot help the many who are poor, it cannot save the few who are rich.

To our sister republics south of our border, we offer a special pledge -- to convert our good words into good deeds -- in a new alliance for progress -- to assist free men and free governments in casting off the chains of poverty. But this peaceful revolution of hope cannot become the prey of hostile powers. Let all our neighbors know that we shall join with them to oppose aggression or subversion anywhere in the Americas. And let every other power know that this Hemisphere intends to remain the master of its own house.

To that world assembly of sovereign states, the United Nations, our last best hope in an age where the instruments of war have far outpaced the instruments of peace, we renew our pledge of support -- to prevent it from becoming merely a forum for invective -- to strengthen its shield of the new and the weak -- and to enlarge the area in which its writ may run.

Finally, to those nations who would make themselves our adversary, we offer not a pledge but a request: that both sides begin anew the quest for peace, before the dark powers of destruction unleashed by science engulf all humanity in planned or accidental self-destruction.

We dare not tempt them with weakness. For only when our arms are sufficient beyond doubt can we be certain beyond doubt that they will never be employed.

But neither can two great and powerful groups of nations take comfort from our present course -- both sides overburdened by the cost of modern weapons, both rightly alarmed by the steady spread of the deadly atom, yet both racing to alter that uncertain balance of terror that stays the hand of mankind's final war.

So let us begin anew -- remembering on both sides that civility is not a sign of weakness, and sincerity is always subject to proof. Let us never negotiate out of fear. But let us never fear to negotiate.

Let both sides explore what problems unite us instead of belaboring those problems which divide us.

Let both sides, for the first time, formulate serious and precise proposals for the inspection and control of arms -- and bring the absolute power to destroy other nations under the absolute control of all nations.

Let both sides seek to invoke the wonders of science instead of its terrors. Together let us explore the stars, conquer the deserts, eradicate disease, tap the ocean depths, and encourage the arts and commerce.

Let both sides unite to heed in all corners of the earth the command of Isaiah -- to "undo the heavy burdens ... and to let the oppressed go free."

And if a beachhead of cooperation may push back the jungle of suspicion, let both sides join in creating a new endeavor, not a new balance of power, but a new world of law, where the strong are just and the weak secure and the peace preserved.

All this will not be finished in the first 100 days. Nor will it be finished in the first 1,000 days, nor in the life of this Administration, nor even perhaps in our lifetime on this planet. But let us begin.

In your h

ands, my fellow citizens, more than in mine, will rest the final success or failure of our course. Since this country was founded, each generation of Americans has been summoned to give testimony to its national loyalty. The graves of young Americans who answered the call to service surround the globe.\n\nNow the trumpet summons us again -- not as a call to bear arms, though arms we need; not as a call to battle, though embattled we are -- but a call to bear the burden of a long twilight struggle, year in and year out, "rejoicing in hope, patient in tribulation" -- a struggle against the common enemies of man: tyranny, poverty, disease, and war itself.\n\nCan we forge against these enemies a grand and global alliance, North and South, East and West, that can assure a more fruitful life for all mankind? Will you join in that historic effort?\n\nIn the long history of the world, only a few generations have been granted the role of defending freedom in its hour of maximum danger. I do not shrink from this responsibility -- I welcome it. I do not believe that any of us would exchange places with any other people or any other generation. The energy, the faith, the devotion which we bring to this endeavor will light our country and all who serve it -- and the glow from that fire can truly light the world.\n\nAnd so, my fellow Americans: ask not what your country can do for you -- ask what you can do for your country.\n\nMy fellow citizens of the world: ask not what America will do for you, but what together we can do for the freedom of man.\n\nFinally, whether you are citizens of America or citizens of the world, ask of us the same high standards of strength and sacrifice which we ask of you. With a good conscience our only sure reward, with history the final judge of our deeds, let us go forth to lead the land we love, asking His blessing and His help, but knowing that here on earth God's work must truly be our own.\n'

In [108...

#raw file of Nixon

Nixon_raw

Out[108...

'Mr. Vice President, Mr. Speaker, Mr. Chief Justice, Senator Cook, Mrs. Eisenhower, and my fellow citizens of this great and good country we share together:\n\nWhen we met here four years ago, America was bleak in spirit, depressed by the prospect of seemingly endless war abroad and of destructive conflict at home.\n\nAs we meet here today, we stand on the threshold of a new era of peace in the world.\n\nThe central question before us is: How shall we use that peace? Let us resolve that this era we are about to enter will not be what other postwar periods have so often been: a time of retreat and isolation that leads to stagnation at home and invites new danger abroad.\n\nLet us resolve that this will be what it can become: a time of great responsibilities greatly borne, in which we renew the spirit and the promise of America as we enter our third century as a nation.\n\nThis past year saw far-reaching results from our new policies for peace. By continuing to revitalize our traditional friendships, and by our missions to Peking and to Moscow, we were able to establish the base for a new and more durable pattern of relationships among the nations of the world. Because of America's bold initiatives, 1972 will be long remembered as the year of the greatest progress since the end of World War II toward a lasting peace in the world.\n\nThe peace we seek in the world is not the flimsy peace which is merely an interlude between wars, but a peace which can endure for generations to come.\n\nIt is important that we understand both the necessity and the limitations of America's role in maintaining that peace.\n\nUnless we in America work to preserve the peace, there will be no peace.\n\nUnless we in America work to preserve freedom, there will be no freedom.\n\nBut let us clearly understand the new nature of America's role, as a result of the new policies we have adopted over these past four years.\n\nWe shall respect our treaty commitments.\n\nWe shall support vigorously the principle that no country has the right to impose its will or rule on another by force.\n\nWe shall continue, in this era of negotiation, to work for the limitation of nuclear arms, and to reduce the danger of confrontation between the great powers.\n\nWe shall do our share in defending peace and freedom in the world. But we shall expect others to do their share.\n\nThe time has passed when America will make every other nation's conflict our own, or make every other nation's future our responsibility, or presume to tell the people of other nations how to manage their own affairs.\n\nJust as we respect the right of each nation to determine its own future, we also recognize the responsibility of each nation to secure its own future.\n\nJust as America's role is indispensable in preserving the world

's peace, so is each nation's role indispensable in preserving its own peace. To get her with the rest of the world, let us resolve to move forward from the beginnings we have made. Let us continue to bring down the walls of hostility which have divided the world for too long, and to build in their place bridges of understanding -- so that despite profound differences between systems of government, the people of the world can be friends. Let us build a structure of peace in the world in which the weak are as safe as the strong -- in which each respects the right of the other to live by a different system -- in which those who would influence others will do so by the strength of their ideas, and not by the force of their arms. Let us accept that high responsibility not as a burden, but gladly -- gladly because the chance to build such a peace is the noblest endeavor in which a nation can engage; gladly, also, because only if we act greatly in meeting our responsibilities abroad will we remain a great Nation, and only if we remain a great Nation will we act greatly in meeting our challenges at home. We have the chance today to do more than ever before in our history to make life better in America -- to ensure better education, better health, better housing, better transportation, a cleaner environment -- to restore respect for law, to make our communities more livable -- and to insure the God-given right of every American to full and equal opportunity. Because the range of our needs is so great -- because the reach of our opportunities is so great -- let us be bold in our determination to meet those needs in new ways. Just as building a structure of peace abroad has required turning away from old policies that failed, so building a new era of progress at home requires turning away from old policies that have failed. Abroad, the shift from old policies to new has not been a retreat from our responsibilities, but a better way to peace. And at home, the shift from old policies to new will not be a retreat from our responsibilities, but a better way to progress. Abroad and at home, the key to those new responsibilities lies in the placing and the division of responsibility. We have lived too long with the consequences of attempting to gather all power and responsibility in Washington. Abroad and at home, the time has come to turn away from the condescending policies of paternalism -- of "Washington knows best." A person can be expected to act responsibly only if he has responsibility. This is human nature. So let us encourage individuals at home and nations abroad to do more for themselves, to decide more for themselves. Let us locate responsibility in more places. Let us measure what we will do for others by what they will do for themselves. That is why today I offer no promise of a purely governmental solution for every problem. We have lived too long with that false promise. In trusting too much in government, we have asked of it more than it can deliver. This leads only to inflated expectations, to reduced individual effort, and to a disappointment and frustration that erode confidence both in what government can do and in what people can do. Government must learn to take less from people so that people can do more for themselves. Let us remember that America was built not by government, but by people -- not by welfare, but by work -- not by shirking responsibility, but by seeking responsibility. In our own lives, let each of us ask -- not just what will government do for me, but what can I do for myself? In the challenges we face together, let each of us ask -- not just how can government help, but how can I help? Your National Government has a great and vital role to play. And I pledge to you that where this Government should act, we will act boldly and we will lead boldly. But just as important is the role that each and every one of us must play, as an individual and as a member of his own community. From this day forward, let each of us make a solemn commitment in his own heart: to bear his responsibility, to do his part, to live his ideals -- so that together, we can see the dawn of a new age of progress for America, and together, as we celebrate our 200th anniversary as a nation, we can do so proud in the fulfillment of our promise to ourselves and to the world. As America's longest and most difficult war comes to an end, let us again learn to debate our differences with civility and decency. And let each of us reach out for that one precious quality government cannot provide -- a new level of respect for the rights and feelings of one another, a new level of respect for the individual human dignity which is the cherished birthright of every American. Above all else, the time has come for us to renew our faith in ourselves and in America. In recent years, that faith has been challenged. Our children have been taught to be ashamed of their country, ashamed of their parents, ashamed of America's record at home and of its role in the world. At every turn, we have been beset by those who find everything wrong with America and little that is right. But I am confident that this will not be the judgment of history on these remarkable times in which we are privileged to live. America's record in this century

y has been unparalleled in the world's history for its responsibility, for its generosity, for its creativity and for its progress.\n\nLet us be proud that our system has produced and provided more freedom and more abundance, more widely shared, than any other system in the history of the world.\n\nLet us be proud that in each of the four wars in which we have been engaged in this century, including the one we are now bringing to an end, we have fought not for our selfish advantage, but to help others resist aggression.\n\nLet us be proud that by our bold, new initiatives, and by our steadfastness for peace with honor, we have made a break-through toward creating in the world what the world has not known before -- a structure of peace that can last, not merely for our time, but for generations to come.\n\nWe are embarking here today on an era that presents challenges great as those any nation, or any generation, has ever faced.\n\nWe shall answer to God, to history, and to our conscience for the way in which we use these years.\n\nAs I stand in this place, so hallowed by history, I think of others who have stood here before me. I think of the dreams they had for America, and I think of how each recognized that he needed help far beyond himself in order to make those dreams come true.\n\nToday, I ask your prayers that in the years ahead I may have God's help in making decisions that are right for America, and I pray for your help so that together we may be worthy of our challenge.\n\nLet us pledge together to make these next four years the best four years in America's history, so that on its 200th birthday America will be as young and as vital as when it began, and as bright a beacon of hope for all the world.\n\nLet us go forward from here confident in hope, strong in our faith in one another, sustained by our faith in God who created us, and striving always to serve His purpose.\n'

Now, let's check length of all 3 files

In [109]...

```
print('Length of Roosevelt_raw:', len(Roosevelt_raw))
print()
print('Length of Kennedy_raw:', len(Kennedy_raw))
print()
print('Length of Nixon_raw:', len(Nixon_raw))
```

Length of Roosevelt_raw: 7571

Length of Kennedy_raw: 7618

Length of Nixon_raw: 9991

The variable 'Roosevelt_raw' contains a string with 7571 characters, Kennedy_raw contains 7618 characters, and Nixon_raw contains 9991 characters. This is the raw content of the speech, including many details we are not interested in such as whitespace, line breaks and blank lines. Notice the \r and \n in the opening line of the file, which is how Python displays the special carriage return and line feed characters. For our language processing, we want to break up the string into words and punctuation. This step is called tokenization, and it produces our familiar structure, a list of words and punctuation.

Removing unwanted characters

In [110]...

```
unwanted_chars = [';', ':', '!', "*", '._', '``', '"""']

# using replace() to
# remove bad_chars
for i in unwanted_chars :
    Roosevelt_raw = Roosevelt_raw.replace(i, '')
```

In [111]...

```
for i in unwanted_chars :
```

```
Kennedy_raw = Kennedy_raw.replace(i, '')
```

```
In [112... for i in unwanted_chars :
            Nixon_raw = Nixon_raw.replace(i, '')
```

```
In [113... # Removing additional space from string using re.sub()

import re

extra_space_remove_R = re.sub(' +', ' ', Roosevelt_raw)
extra_space_remove_K = re.sub(' +', ' ', Kennedy_raw)
extra_space_remove_N = re.sub(' +', ' ', Nixon_raw)
```

```
In [114... #removing punctuations

re.sub("[^9A-Za-z ]", "", Roosevelt_raw)
re.sub("[^9A-Za-z ]", "", Kennedy_raw)
re.sub("[^9A-Za-z ]", "", Nixon_raw)
```

```
Out[114... 'Mr Vice President Mr Speaker Mr Chief Justice Senator Cook Mrs Eisenhower and my fellow
citizens of this great and good country we share togetherWhen we met here four years ago
America was bleak in spirit depressed by the prospect of seemingly endless war abroad an
d of destructive conflict at homeAs we meet here today we stand on the threshold of a ne
w era of peace in the worldThe central question before us is How shall we use that peace
Let us resolve that this era we are about to enter will not be what other postwar period
s have so often been a time of retreat and isolation that leads to stagnation at home an
d invites new danger abroadLet us resolve that this will be what it can become a time of
great responsibilities greatly borne in which we renew the spirit and the promise of Ame
rica as we enter our third century as a nationThis past year saw far-reaching results fr
om our new policies for peace By continuing to revitalize our traditional friendships an
d by our missions to Peking and to Moscow we were able to establish the base for a new a
nd more durable pattern of relationships among the nations of the world Because of Ameri
cas bold initiatives 9 will be long remembered as the year of the greatest progress sinc
e the end of World War II toward a lasting peace in the worldThe peace we seek in the wo
rld is not the flimsy peace which is merely an interlude between wars but a peace which
can endure for generations to comeIt is important that we understand both the necessity
and the limitations of Americas role in maintaining that peaceUnless we in America work
to preserve the peace there will be no peaceUnless we in America work to preserve freedo
m there will be no freedomBut let us clearly understand the new nature of Americas role
as a result of the new policies we have adopted over these past four yearsWe shall respe
ct our treaty commitmentsWe shall support vigorously the principle that no country has t
he right to impose its will or rule on another by forceWe shall continue in this era of
negotiation to work for the limitation of nuclear arms and to reduce the danger of confr
ontation between the great powersWe shall do our share in defending peace and freedom in
the world But we shall expect others to do their shareThe time has passed when America w
ill make every other nations conflict our own or make every other nations future our res
ponsibility or presume to tell the people of other nations how to manage their own affai
rsJust as we respect the right of each nation to determine its own future we also recogni
ze the responsibility of each nation to secure its own futureJust as Americas role is i
ndispensable in preserving the worlds peace so is each nations role indispensable in pre
serving its own peaceTogether with the rest of the world let us resolve to move forward
from the beginnings we have made Let us continue to bring down the walls of hostility wh
ich have divided the world for too long and to build in their place bridges of understand
ing so that despite profound differences between systems of government the people of t
he world can be friendsLet us build a structure of peace in the world in which the weak
are as safe as the strong in which each respects the right of the other to live by a di
```

fferent system in which those who would influence others will do so by the strength of their ideas and not by the force of their arms. Let us accept that high responsibility not as a burden but gladly, gladly because the chance to build such a peace is the noblest endeavor in which a nation can engage, gladly also because only if we act greatly in meeting our responsibilities abroad will we remain a great Nation and only if we remain a great Nation will we act greatly in meeting our challenges at home. We have the chance today to do more than ever before in our history to make life better in America, to ensure better education, better health, better housing, better transportation, a cleaner environment, to restore respect for law, to make our communities more livable, and to insure the God-given right of every American to full and equal opportunity. Because the range of our needs is so great, because the reach of our opportunities is so great, let us be bold in our determination to meet those needs in new ways. Just as building a structure of peace abroad has required turning away from old policies that failed, so building a new era of progress at home requires turning away from old policies that have failed. Abroad, the shift from old policies to new has not been a retreat from our responsibilities but a better way to peace. And at home, the shift from old policies to new will not be a retreat from our responsibilities but a better way to progress. Abroad and at home, the key to those new responsibilities lies in the placing and the division of responsibility. We have lived too long with the consequences of attempting to gather all power and responsibility in Washington. Abroad and at home, the time has come to turn away from the condescending policies of paternalism. Of Washington, we know best. A person can be expected to act responsibly only if he has responsibility. This is human nature. So let us encourage individuals at home and nations abroad to do more for themselves, to decide more for themselves. Let us locate responsibility in more places. Let us measure what we will do for others by what they will do for themselves. That is why today I offer no promise of a purely governmental solution for every problem. We have lived too long with that false promise. In trusting too much in government, we have asked of it more than it can deliver. This leads only to inflated expectations, to reduced individual effort, and to a disappointment and frustration that erode confidence both in what government can do and in what people can do. Government must learn to take less from people so that people can do more for themselves. Let us remember that America was built not by government but by people, not by welfare but by work, not by shirking responsibility but by seeking responsibility. In our own lives, let each of us ask, not just what will government do for me, but what can I do for myself. In the challenges we face together, let each of us ask, not just how can government help, but how can I help. Your National Government has a great and vital role to play. And I pledge to you that where this Government should act, we will act boldly, and we will lead boldly. But just as important is the role that each and every one of us must play as an individual and as a member of his own community. From this day forward, let each of us make a solemn commitment in his own heart to bear his responsibility to do his part to live his ideals, so that together we can see the dawn of a new age of progress for America, and together as we celebrate our 75th anniversary as a nation, we can do so proud in the fulfillment of our promise to ourselves and to the world. As America's longest and most difficult war comes to an end, let us again learn to debate our differences with civility and decency. And let each of us reach out for that one precious quality government cannot provide: a new level of respect for the rights and feelings of one another, a new level of respect for the individual human dignity which is the cherished birthright of every American. Above all else, the time has come for us to renew our faith in ourselves and in America. In recent years, that faith has been challenged. Our children have been taught to be ashamed of their country, ashamed of their parents, ashamed of America's record at home and of its role in the world. At every turn, we have been beset by those who find everything wrong with America and little that is right. But I am confident that this will not be the judgment of history on these remarkable times in which we are privileged to live. America's record in this century has been unparalleled in the world's history for its responsibility, for its generosity, for its creativity, and for its progress. Let us be proud that our system has produced and provided more freedom and more abundance, more widely shared than any other system in the history of the world. Let us be proud that in each of the four wars in which we have been engaged in this century, including the one we are now bringing to an end, we have fought not for our selfish advantage but to help others resist aggression. Let us be proud that by our bold new initiatives and by our steadfastness for peace with honor, we have made a break-through toward creating in the world what the world has not known before: a structure of peace that can last not merely for our time but for generations to come. We are embarking here today on a

n era that presents challenges great as those any nation or any generation has ever face
 dWe shall answer to God to history and to our conscience for the way in which we use the
 se yearsAs I stand in this place so hallowed by history I think of others who have stood
 here before me I think of the dreams they had for America and I think of how each recogn
 ized that he needed help far beyond himself in order to make those dreams come trueToday
 I ask your prayers that in the years ahead I may have Gods help in making decisions that
 are right for America and I pray for your help so that together we may be worthy of our
 challengeLet us pledge together to make these next four years the best four years in Ame
 ricas history so that on its th birthday America will be as young and as vital as when i
 t began and as bright a beacon of hope for all the worldLet us go forward from here conf
 ident in hope strong in our faith in one another sustained by our faith in God who creat
 ed us and striving always to serve His purpose'

In [115...

```
#case normalization(all upper case to lower case conversion)

Roosevelt_raw = Roosevelt_raw.lower()
Kennedy_raw = Kennedy_raw.lower()
Nixon_raw = Nixon_raw.lower()
```

In [116...

```
#Again Checking Length

print('Length of Roosevelt_raw:', len(Roosevelt_raw))
print()
print('Length of Kennedy_raw:', len(Kennedy_raw))
print()
print('Length of Nixon_raw:', len(Nixon_raw))
```

Length of Roosevelt_raw: 7515

Length of Kennedy_raw: 7563

Length of Nixon_raw: 9951

Now let's check number of sentences in all speeches

tokenization

In [117...

```
R_tokens = nltk.word_tokenize(Roosevelt_raw)

print('length of tokens:',len(R_tokens))

print(R_tokens[:10]) #print top 10 tokens

print('-----')

K_tokens = nltk.word_tokenize(Kennedy_raw)

print('length of tokens:',len(K_tokens))

print(K_tokens[:10]) #print top 10 tokens

print('-----')

N_tokens = nltk.word_tokenize(Nixon_raw)

print('length of tokens:',len(N_tokens))
```

```
print(N_tokens[:10]) #print top 10 tokens
```

length of tokens: 1492

```
['on', 'each', 'national', 'day', 'of', 'inauguration', 'since', '1789', ',', 'the']
```

length of tokens: 1512

```
['vice', 'president', 'johnson', ',', 'mr.', 'speaker', ',', 'mr.', 'chief', 'justice']
```

length of tokens: 1983

```
['mr.', 'vice', 'president', ',', 'mr.', 'speaker', ',', 'mr.', 'chief', 'justice']
```

In [118...

```
#Number of sentences in Roosevelt's speech
```

```
R_sentences= nltk.sent_tokenize(Roosevelt_raw)
```

```
R_length= len(R_sentences)
```

```
print("Number of sentences in Roosevelt's speech:",R_length)
```

Number of sentences in Roosevelt's speech: 68

In [119...

```
#Number of sentences in Kennedy's speech
```

```
K_sentences= nltk.sent_tokenize(Kennedy_raw)
```

```
K_length= len(K_sentences)
```

```
print("Number of sentences in Kennedy's speech:",K_length)
```

Number of sentences in Kennedy's speech: 52

In [120...

```
#Number of sentences in Nixon's speech
```

```
N_sentences= nltk.sent_tokenize(Nixon_raw)
```

```
N_length= len(N_sentences)
```

```
print("Number of sentences in Nixon's speech:",N_length)
```

Number of sentences in Nixon's speech: 68

We will use `nltk.Text(tokens)` function on the tokens and further do more linguistic processing using text.

In [121...

```
R_text = nltk.Text(R_tokens)
```

In [122...

```
K_text = nltk.Text(K_tokens)
```

In [123...

```
N_text = nltk.Text(N_tokens)
```

```
type(N_text)
```

Out[123... `nltk.text.Text`

Checking some random texts from all 3

In [124...

```
print('Checking texts from R_text:\n\n',R_text[820:835])
print('-----\n')
print('Checking texts from K_text:\n\n',K_text[700:720])
print('-----\n')
print('Checking texts from N_text:\n\n',N_text[900:910])
print()
```

Checking texts from R_text:

```
['the', 'democratic', 'aspiration', 'is', 'no', 'mere', 'recent', 'phase', 'in', 'human', 'history', '.', 'it', 'is', 'human']
-----
```

Checking texts from K_text:

```
['area', 'in', 'which', 'its', 'writ', 'may', 'run', '.', 'finally', ',', 'to', 'those', 'nations', 'who', 'would', 'make', 'themselves', 'our', 'adversary', ',']
-----
```

Checking texts from N_text:

```
['to', 'new', 'has', 'not', 'been', 'a', 'retreat', 'from', 'our', 'responsibilities']
```

2.2 Remove all the stopwords from all three speeches.

In [125...

```
#Below are the common stopwords

from nltk.corpus import stopwords

print(stopwords.words('english'))
```

```
['i', 'me', 'my', 'myself', 'we', 'our', 'ours', 'ourselves', 'you', "you're", "you've", "you'll", "you'd", 'your', 'yours', 'yourself', 'yourselves', 'he', 'him', 'his', 'himself', 'she', "she's", 'her', 'hers', 'herself', 'it', "it's", 'its', 'itself', 'they', 'them', 'their', 'theirs', 'themselves', 'what', 'which', 'who', 'whom', 'this', 'that', "that'll", 'these', 'those', 'am', 'is', 'are', 'was', 'were', 'be', 'been', 'being', 'have', 'has', 'had', 'having', 'do', 'does', 'did', 'doing', 'a', 'an', 'the', 'and', 'but', 'if', 'or', 'because', 'as', 'until', 'while', 'of', 'at', 'by', 'for', 'with', 'about', 'against', 'between', 'into', 'through', 'during', 'before', 'after', 'above', 'below', 'to', 'from', 'up', 'down', 'in', 'out', 'on', 'off', 'over', 'under', 'again', 'further', 'then', 'once', 'here', 'there', 'when', 'where', 'why', 'how', 'all', 'any', 'both', 'each', 'few', 'more', 'most', 'other', 'some', 'such', 'no', 'nor', 'not', 'only', 'own', 'same', 'so', 'than', 'too', 'very', 's', 't', 'can', 'will', 'just', 'don', "don't", 'should', "should've", 'now', 'd', 'll', 'm', 'o', 're', 've', 'y', 'ain', 'aren', "aren't", 'couldn', "couldn't", 'didn', "didn't", 'doesn', "doesn't", 'hadn', "hadn't", 'hasn', "hasn't", 'haven', "haven't", 'isn', "isn't", 'ma', 'mightn', "mightn't", 'mustn', "mustn't", 'needn', "needn't", 'shan', "shan't", 'shouldn', "shouldn't", 'wasn', "wasn't", 'weren', "weren't", 'won', "won't", 'wouldn', "wouldn't"]
```

In [126...

```
#Removing the stopwords from Roosevelt speech(R_text)

#R_tokens is tokenization for Roosevelt

stop_words = set(stopwords.words('english') + list(string.punctuation))
```

```

Roosevelt_without_stopwords_punc = [w for w in R_tokens if w.lower() not in stop_words]

Roosevelt_without_stopwords_punc = []

for w in R_tokens:
    if w not in stop_words:
        Roosevelt_without_stopwords_punc.append(w)

print(Roosevelt_without_stopwords_punc)

```

```

['national', 'day', 'inauguration', 'since', '1789', 'people', 'renewed', 'sense', 'dedi
cation', 'united', 'states', 'washington', "'s", 'day', 'task', 'people', 'create', 'wel
d', 'together', 'nation', 'lincoln', "'s", 'day', 'task', 'people', 'preserve', 'natio
n', 'disruption', 'within', 'day', 'task', 'people', 'save', 'nation', 'institutions',
'disruption', 'without', 'us', 'come', 'time', 'midst', 'swift', 'happenings', 'pause',
'moment', 'take', 'stock', 'recall', 'place', 'history', 'rediscover', 'may', 'risk', 'r
eal', 'peril', 'inaction', 'lives', 'nations', 'determined', 'count', 'years', 'lifetim
e', 'human', 'spirit', 'life', 'man', 'three-score', 'years', 'ten', 'little', 'little',
'less', 'life', 'nation', 'fullness', 'measure', 'live', 'men', 'doubt', 'men', 'believ
e', 'democracy', 'form', 'government', 'frame', 'life', 'limited', 'measured', 'kind',
'mystical', 'artificial', 'fate', 'unexplained', 'reason', 'tyranny', 'slavery', 'becom
e', 'surging', 'wave', 'future', 'freedom', 'ebbing', 'tide', 'americans', 'know', 'tru
e', 'eight', 'years', 'ago', 'life', 'republic', 'seemed', 'frozen', 'fatalistic', 'terr
or', 'proved', 'true', 'midst', 'shock', 'acted', 'acted', 'quickly', 'boldly', 'decisiv
ely', 'later', 'years', 'living', 'years', 'fruitful', 'years', 'people', 'democracy',
'brought', 'us', 'greater', 'security', 'hope', 'better', 'understanding', 'life', "'s",
'ideals', 'measured', 'material', 'things', 'vital', 'present', 'future', 'experience',
'democracy', 'successfully', 'survived', 'crisis', 'home', 'put', 'away', 'many', 'evi
l', 'things', 'built', 'new', 'structures', 'enduring', 'lines', 'maintained', 'fact',
'democracy', 'action', 'taken', 'within', 'three-way', 'framework', 'constitution', 'uni
ted', 'states', 'coordinate', 'branches', 'government', 'continue', 'freely', 'functio
n', 'bill', 'rights', 'remains', 'inviolable', 'freedom', 'elections', 'wholly', 'maintai
ned', 'prophets', 'downfall', 'american', 'democracy', 'seen', 'dire', 'predictions', 'c
ome', 'naught', 'democracy', 'dying', 'know', 'seen', 'reviveand', 'grow', 'know', 'di
e', 'built', 'unhampered', 'initiative', 'individual', 'men', 'women', 'joined', 'togeth
er', 'common', 'enterprise', 'enterprise', 'undertaken', 'carried', 'free', 'expressio
n', 'free', 'majority', 'know', 'democracy', 'alone', 'forms', 'government', 'enlists',
'full', 'force', 'men', "'s", 'enlightened', 'know', 'democracy', 'alone', 'constructe
d', 'unlimited', 'civilization', 'capable', 'infinite', 'progress', 'improvement', 'huma
n', 'life', 'know', 'look', 'surface', 'sense', 'still', 'spreading', 'every', 'continen
t', 'humane', 'advanced', 'end', 'unconquerable', 'forms', 'human', 'society', 'nation',
'like', 'person', 'bodya', 'body', 'must', 'fed', 'clothed', 'housed', 'invigorated', 'r
ested', 'manner', 'measures', 'objectives', 'time', 'nation', 'like', 'person', 'mind',
'mind', 'must', 'kept', 'informed', 'alert', 'must', 'know', 'understands', 'hopes', 'ne
eds', 'neighbors', 'nations', 'live', 'within', 'narrowing', 'circle', 'world', 'natio
n', 'like', 'person', 'something', 'deeper', 'something', 'permanent', 'something', 'lar
ger', 'sum', 'parts', 'something', 'matters', 'future', 'calls', 'forth', 'sacred', 'gua
rding', 'present', 'thing', 'find', 'difficult', 'even', 'impossible', 'hit', 'upon', 's
ingle', 'simple', 'word', 'yet', 'understand', 'spirit', 'faith', 'america', 'product',
'centuries', 'born', 'multitudes', 'came', 'many', 'lands', 'high', 'degree', 'mostly',
'plain', 'people', 'sought', 'early', 'late', 'find', 'freedom', 'freely', 'democratic',
'aspiration', 'mere', 'recent', 'phase', 'human', 'history', 'human', 'history', 'permea
ted', 'ancient', 'life', 'early', 'peoples', 'blazed', 'anew', 'middle', 'ages', 'writte
n', 'magna', 'charta', 'americas', 'impact', 'irresistible', 'america', 'new', 'world',
'tongues', 'peoples', 'continent', 'new-found', 'land', 'came', 'believed', 'could', 'cr
eate', 'upon', 'continent', 'new', 'life', 'life', 'new', 'freedom', 'vitality', 'writte
n', 'mayflower', 'compact', 'declaration', 'independence', 'constitution', 'united', 'st
ates', 'gettysburg', 'address', 'first', 'came', 'carry', 'longings', 'spirit', 'million
s', 'followed', 'stock', 'sprang', 'moved', 'forward', 'constantly', 'consistently', 'to
ward', 'ideal', 'gained', 'stature', 'clarity', 'generation', 'hopes', 'republic', 'fore
ver', 'tolerate', 'either', 'undeserved', 'poverty', 'self-serving', 'wealth', 'know',

```

```
'still', 'far', 'go', 'must', 'greatly', 'build', 'security', 'opportunity', 'knowledg
e', 'every', 'citizen', 'measure', 'justified', 'resources', 'capacity', 'land', 'enoug
h', 'achieve', 'purposes', 'alone', 'enough', 'clothe', 'feed', 'body', 'nation', 'instr
uct', 'inform', 'mind', 'also', 'spirit', 'three', 'greatest', 'spirit', 'without', 'bod
y', 'mind', 'men', 'know', 'nation', 'could', 'live', 'spirit', 'america', 'killed', 'ev
en', 'though', 'nation', '"s", 'body', 'mind', 'constricted', 'alien', 'world', 'lived',
'america', 'know', 'would', 'perished', 'spirit', 'faith', 'speaks', 'us', 'daily', 'liv
es', 'ways', 'often', 'unnoticed', 'seem', 'obvious', 'speaks', 'us', 'capital', 'natio
n', 'speaks', 'us', 'processes', 'governing', 'sovereignties', '48', 'states', 'speaks',
'us', 'counties', 'cities', 'towns', 'villages', 'speaks', 'us', 'nations', 'hemispher
e', 'across', 'seas', 'enslaved', 'well', 'free', 'sometimes', 'fail', 'hear', 'heed',
'voices', 'freedom', 'us', 'privilege', 'freedom', 'old', 'old', 'story', 'destiny', 'am
erica', 'proclaimed', 'words', 'prophecy', 'spoken', 'first', 'president', 'first', 'ina
ugural', '1789', 'words', 'almost', 'directed', 'would', 'seem', 'year', '1941', '`',
'preservation', 'sacred', 'fire', 'liberty', 'destiny', 'republican', 'model', 'governme
nt', 'justly', 'considered', 'deeply', 'finally', 'staked', 'experiment', 'intrusted',
'hands', 'american', 'people', '"', 'lose', 'sacred', 'fireif', 'let', 'smothered', 'do
ubt', 'fear', 'shall', 'reject', 'destiny', 'washington', 'strove', 'valiantly', 'triump
hantly', 'establish', 'preservation', 'spirit', 'faith', 'nation', 'furnish', 'highest',
'justification', 'every', 'sacrifice', 'may', 'make', 'cause', 'national', 'defense', 'f
ace', 'great', 'perils', 'never', 'encountered', 'strong', 'purpose', 'protect', 'perpet
uate', 'integrity', 'democracy', 'muster', 'spirit', 'america', 'faith', 'america', 'ret
reat', 'content', 'stand', 'still', 'americans', 'go', 'forward', 'service', 'country',
'god']
```

```
In [127... len(Roosevelt_without_stopwords_punc)
```

```
Out[127... 632
```

```
In [128... #Removing the stopwords from Kennedy speech

#K_tokens is tokenization for Kennedy

stop_words = set(stopwords.words('english') +list(string.punctuation))

Kennedy_without_stopwords_punc = [w for w in K_tokens if w.lower() not in stop_words]

Kennedy_without_stopwords_punc = []

for w in K_tokens:
    if w not in stop_words:
        Kennedy_without_stopwords_punc.append(w)

print(Kennedy_without_stopwords_punc)
```

```
['vice', 'president', 'johnson', 'mr.', 'speaker', 'mr.', 'chief', 'justice', 'presiden
t', 'eisenhower', 'vice', 'president', 'nixon', 'president', 'truman', 'reverend', 'cler
gy', 'fellow', 'citizens', 'observe', 'today', 'victory', 'party', 'celebration', 'freed
om', 'symbolizing', 'end', 'well', 'beginning', 'signifying', 'renewal', 'well', 'chang
e', 'sworn', 'almighty', 'god', 'solemn', 'oath', 'forebears', 'l', 'prescribed', 'nearl
y', 'century', 'three', 'quarters', 'ago', 'world', 'different', 'man', 'holds', 'morta
l', 'hands', 'power', 'abolish', 'forms', 'human', 'poverty', 'forms', 'human', 'life',
'yet', 'revolutionary', 'beliefs', 'forebears', 'fought', 'still', 'issue', 'around', 'g
lobe', 'belief', 'rights', 'man', 'come', 'generosity', 'state', 'hand', 'god', 'dare',
'forget', 'today', 'heirs', 'first', 'revolution', 'let', 'word', 'go', 'forth', 'time',
'place', 'friend', 'foe', 'alike', 'torch', 'passed', 'new', 'generation', 'americans',
'born', 'century', 'tempered', 'war', 'disciplined', 'hard', 'bitter', 'peace', 'proud',
'ancient', 'heritage', 'unwilling', 'witness', 'permit', 'slow', 'undoing', 'human', 'ri
ghts', 'nation', 'always', 'committed', 'committed', 'today', 'home', 'around', 'world',
```

'let', 'every', 'nation', 'know', 'whether', 'wishes', 'us', 'well', 'ill', 'shall', 'pay', 'price', 'bear', 'burden', 'meet', 'hardship', 'support', 'friend', 'oppose', 'foe', 'order', 'assure', 'survival', 'success', 'liberty', 'much', 'pledge', 'old', 'allies', 'whose', 'cultural', 'spiritual', 'origins', 'share', 'pledge', 'loyalty', 'faithful', 'friends', 'united', 'little', 'host', 'cooperative', 'ventures', 'divided', 'little', 'dare', 'meet', 'powerful', 'challenge', 'odds', 'split', 'asunder', 'new', 'states', 'welcome', 'ranks', 'free', 'pledge', 'word', 'one', 'form', 'colonial', 'control', 'shall', 'passed', 'away', 'merely', 'replaced', 'far', 'iron', 'tyranny', 'shall', 'always', 'expect', 'find', 'supporting', 'view', 'shall', 'always', 'hope', 'find', 'strongly', 'supporting', 'freedom', 'remember', 'past', 'foolishly', 'sought', 'power', 'riding', 'back', 'tiger', 'ended', 'inside', 'peoples', 'huts', 'villages', 'across', 'globe', 'struggling', 'break', 'bonds', 'mass', 'misery', 'pledge', 'best', 'efforts', 'help', 'help', 'whatever', 'period', 'required', 'communists', 'may', 'seek', 'votes', 'right', 'free', 'society', 'help', 'many', 'poor', 'save', 'rich', 'sister', 'republics', 'south', 'border', 'offer', 'special', 'pledge', 'convert', 'good', 'words', 'good', 'deeds', 'new', 'alliance', 'progress', 'assist', 'free', 'men', 'free', 'governments', 'casting', 'chains', 'poverty', 'peaceful', 'revolution', 'hope', 'become', 'prey', 'hostile', 'powers', 'let', 'neighbors', 'know', 'shall', 'join', 'oppose', 'aggression', 'subversion', 'anywhere', 'americas', 'let', 'every', 'power', 'know', 'hemisphere', 'intends', 'remain', 'master', 'house', 'world', 'assembly', 'sovereign', 'states', 'united', 'nations', 'last', 'best', 'hope', 'age', 'instruments', 'war', 'far', 'outpaced', 'instruments', 'peace', 'renew', 'pledge', 'supportto', 'prevent', 'becoming', 'merely', 'forum', 'invasive', 'strengthen', 'shield', 'new', 'weak', 'enlarge', 'area', 'writ', 'may', 'run', 'finally', 'nations', 'would', 'make', 'adversary', 'offer', 'pledge', 'request', 'sides', 'begin', 'anew', 'quest', 'peace', 'dark', 'powers', 'destruction', 'unleashed', 'science', 'engulf', 'humanity', 'planned', 'accidental', 'self-destruction', 'dare', 'tempt', 'weakness', 'arms', 'sufficient', 'beyond', 'doubt', 'certain', 'beyond', 'doubt', 'never', 'employed', 'neither', 'two', 'great', 'powerful', 'groups', 'nations', 'take', 'comfort', 'present', 'course', 'sides', 'overburdened', 'cost', 'modern', 'weapons', 'rightly', 'alarmed', 'steady', 'spread', 'deadly', 'atom', 'yet', 'racing', 'alter', 'uncertain', 'balance', 'terror', 'stays', 'hand', 'mankind', '"s"', 'final', 'war', 'let', 'us', 'begin', 'anew', 'remembering', 'sides', 'civility', 'sign', 'weakness', 'sincerity', 'always', 'subject', 'proof', 'let', 'us', 'never', 'negotiate', 'fear', 'let', 'us', 'never', 'fear', 'negotiate', 'let', 'sides', 'explore', 'problems', 'unite', 'us', 'instead', 'belaboring', 'problems', 'divide', 'us', 'let', 'sides', 'first', 'time', 'formulate', 'serious', 'precise', 'proposals', 'inspection', 'control', 'arms', 'bring', 'absolute', 'power', 'destroy', 'nations', 'absolute', 'control', 'nations', 'let', 'sides', 'seek', 'invoke', 'wonders', 'science', 'instead', 'terrors', 'together', 'let', 'us', 'explore', 'stars', 'conquer', 'deserts', 'eradicate', 'disease', 'tap', 'ocean', 'depths', 'encourage', 'arts', 'commerce', 'let', 'sides', 'unite', 'heed', 'corners', 'earth', 'command', 'isaiah', 'undo', 'heavy', 'burdens', '...', 'let', 'oppressed', 'go', 'free', 'beachhead', 'cooperation', 'may', 'push', 'back', 'jungle', 'suspicion', 'let', 'sides', 'join', 'creating', 'new', 'endeavor', 'new', 'balance', 'power', 'new', 'world', 'law', 'strong', 'weak', 'secure', 'peace', 'preserved', 'finished', 'first', '100', 'days', 'finished', 'first', '1,000', 'days', 'life', 'administration', 'even', 'perhaps', 'lifetime', 'planet', 'let', 'us', 'begin', 'hands', 'fellow', 'citizens', 'mine', 'rest', 'final', 'success', 'failure', 'course', 'since', 'country', 'founded', 'generation', 'americans', 'summoned', 'give', 'testimony', 'national', 'loyalty', 'graves', 'young', 'americans', 'answered', 'call', 'service', 'surround', 'globe', 'trumpet', 'summons', 'us', 'call', 'bear', 'arms', 'though', 'arms', 'need', 'call', 'battle', 'though', 'embattled', 'call', 'bear', 'burden', 'long', 'twilight', 'struggle', 'year', 'year', 'rejoicing', 'hope', 'patient', 'tribulation', '"', 'struggle', 'common', 'enemies', 'man', 'tyranny', 'poverty', 'disease', 'war', 'forge', 'enemies', 'grand', 'global', 'alliance', 'north', 'south', 'east', 'west', 'assure', 'fruitful', 'life', 'mankind', 'join', 'historic', 'effort', 'long', 'history', 'world', 'generations', 'granted', 'role', 'defending', 'freedom', 'hour', 'maximum', 'danger', 'shrink', 'responsibility', 'welcome', 'believe', 'us', 'would', 'exchange', 'places', 'people', 'generation', 'energy', 'faith', 'devotion', 'bring', 'endeavor', 'light', 'country', 'serve', 'glow', 'fire', 'truly', 'light', 'world', 'fellow', 'americans', 'ask', 'country', 'ask', 'country', 'fellow', 'citizens', 'world', 'ask', 'america', 'together', 'freedom', 'man', 'finally', 'whether', 'citizens', 'america', 'citizens', 'world', 'ask', 'us', 'h

```
igh', 'standards', 'strength', 'sacrifice', 'ask', 'good', 'conscience', 'sure', 'reward', 'history', 'final', 'judge', 'deeds', 'let', 'us', 'go', 'forth', 'lead', 'land', 'love', 'asking', 'blessing', 'help', 'knowing', 'earth', 'god', "'s", 'work', 'must', 'truly']
```

```
In [129... len(Kennedy_without_stopwords_punc)
```

```
Out[129... 696
```

```
In [130... #Removing the stopwords from Nixon speech
```

```
#N_tokens is tokenization for Nixon
```

```
stop_words = set(stopwords.words('english') +list(string.punctuation))
```

```
Nixon_without_stopwords_punc = [w for w in N_tokens if w.lower() not in stop_words]
```

```
Nixon_without_stopwords_punc = []
```

```
for w in N_tokens:
    if w not in stop_words:
        Nixon_without_stopwords_punc.append(w)
```

```
print(Nixon_without_stopwords_punc)
print('-----')
print(N_tokens)
```

```
['mr.', 'vice', 'president', 'mr.', 'speaker', 'mr.', 'chief', 'justice', 'senator', 'co
ok', 'mrs.', 'eisenhower', 'fellow', 'citizens', 'great', 'good', 'country', 'share', 't
ogether', 'met', 'four', 'years', 'ago', 'america', 'bleak', 'spirit', 'depressed', 'pro
spect', 'seemingly', 'endless', 'war', 'abroad', 'destructive', 'conflict', 'home', 'mee
t', 'today', 'stand', 'threshold', 'new', 'era', 'peace', 'world', 'central', 'questio
n', 'us', 'shall', 'use', 'peace', 'let', 'us', 'resolve', 'era', 'enter', 'postwar', 'p
eriods', 'often', 'time', 'retreat', 'isolation', 'leads', 'stagnation', 'home', 'invite
s', 'new', 'danger', 'abroad', 'let', 'us', 'resolve', 'become', 'time', 'great', 'respo
nsibilities', 'greatly', 'borne', 'renew', 'spirit', 'promise', 'america', 'enter', 'thi
rd', 'century', 'nation', 'past', 'year', 'saw', 'far-reaching', 'results', 'new', 'poli
cies', 'peace', 'continuing', 'revitalize', 'traditional', 'friendships', 'missions', 'p
eking', 'moscow', 'able', 'establish', 'base', 'new', 'durable', 'pattern', 'relationshi
ps', 'among', 'nations', 'world', 'america', "'s", 'bold', 'initiatives', '1972', 'lon
g', 'remembered', 'year', 'greatest', 'progress', 'since', 'end', 'world', 'war', 'ii',
'toward', 'lasting', 'peace', 'world', 'peace', 'seek', 'world', 'flimsy', 'peace', 'mer
ely', 'interlude', 'wars', 'peace', 'endure', 'generations', 'come', 'important', 'under
stand', 'necessity', 'limitations', 'america', "'s", 'role', 'maintaining', 'peace', 'un
less', 'america', 'work', 'preserve', 'peace', 'peace', 'unless', 'america', 'work', 'pr
eserve', 'freedom', 'freedom', 'let', 'us', 'clearly', 'understand', 'new', 'nature', 'a
merica', "'s", 'role', 'result', 'new', 'policies', 'adopted', 'past', 'four', 'years',
'shall', 'respect', 'treaty', 'commitments', 'shall', 'support', 'vigorously', 'principl
e', 'country', 'right', 'impose', 'rule', 'another', 'force', 'shall', 'continue', 'er
a', 'negotiation', 'work', 'limitation', 'nuclear', 'arms', 'reduce', 'danger', 'confron
tation', 'great', 'powers', 'shall', 'share', 'defending', 'peace', 'freedom', 'world',
'shall', 'expect', 'others', 'share', 'time', 'passed', 'america', 'make', 'every', 'nat
ion', "'s", 'conflict', 'make', 'every', 'nation', "'s", 'future', 'responsibility', 'pr
esume', 'tell', 'people', 'nations', 'manage', 'affairs', 'respect', 'right', 'nation',
'determine', 'future', 'also', 'recognize', 'responsibility', 'nation', 'secure', 'futur
e', 'america', "'s", 'role', 'indispensable', 'preserving', 'world', "'s", 'peace', 'nat
ion', "'s", 'role', 'indispensable', 'preserving', 'peace', 'together', 'rest', 'world',
'let', 'us', 'resolve', 'move', 'forward', 'beginnings', 'made', 'let', 'us', 'continu
e', 'bring', 'walls', 'hostility', 'divided', 'world', 'long', 'build', 'place', 'bridge
```

s', 'understanding', 'despite', 'profound', 'differences', 'systems', 'government', 'people', 'world', 'friends', 'let', 'us', 'build', 'structure', 'peace', 'world', 'weak', 'safe', 'strong', 'respects', 'right', 'live', 'different', 'system', 'would', 'influence', 'others', 'strength', 'ideas', 'force', 'arms', 'let', 'us', 'accept', 'high', 'responsibility', 'burden', 'gladly', 'gladly', 'chance', 'build', 'peace', 'noblest', 'endeavor', 'nation', 'engage', 'gladly', 'also', 'act', 'greatly', 'meeting', 'responsibilities', 'abroad', 'remain', 'great', 'nation', 'remain', 'great', 'nation', 'act', 'greatly', 'meeting', 'challenges', 'home', 'chance', 'today', 'ever', 'history', 'make', 'life', 'better', 'america', 'ensure', 'better', 'education', 'better', 'health', 'better', 'housing', 'better', 'transportation', 'cleaner', 'environment', 'restore', 'respect', 'law', 'make', 'communities', 'livable', 'insure', 'god-given', 'right', 'every', 'american', 'full', 'equal', 'opportunity', 'range', 'needs', 'great', 'reach', 'opportunities', 'great', 'let', 'us', 'bold', 'determination', 'meet', 'needs', 'new', 'ways', 'building', 'structure', 'peace', 'abroad', 'required', 'turning', 'away', 'old', 'policies', 'failed', 'building', 'new', 'era', 'progress', 'home', 'requires', 'turning', 'away', 'old', 'policies', 'failed', 'abroad', 'shift', 'old', 'policies', 'new', 'retreat', 'responsibilities', 'better', 'way', 'peace', 'home', 'shift', 'old', 'policies', 'new', 'retreat', 'responsibilities', 'better', 'way', 'progress', 'abroad', 'home', 'key', 'new', 'responsibilities', 'lies', 'placing', 'division', 'responsibility', 'lived', 'long', 'consequences', 'attempting', 'gather', 'power', 'responsibility', 'washington', 'abroad', 'home', 'time', 'come', 'turn', 'away', 'condescending', 'policies', 'paternalism', 'washington', 'knows', 'best', 'person', 'expected', 'act', 'responsibly', 'responsibility', 'human', 'nature', 'let', 'us', 'encourage', 'individuals', 'home', 'nations', 'abroad', 'decide', 'let', 'us', 'locate', 'responsibility', 'places', 'let', 'us', 'measure', 'others', 'today', 'offer', 'promise', 'purely', 'governmental', 'solution', 'every', 'problem', 'lived', 'long', 'false', 'promise', 'trusting', 'much', 'government', 'asked', 'deliver', 'leads', 'inflated', 'expectations', 'reduced', 'individual', 'effort', 'disappointment', 'frustration', 'erode', 'confidence', 'government', 'people', 'government', 'must', 'learn', 'take', 'less', 'people', 'people', 'let', 'us', 'remember', 'america', 'built', 'government', 'people', 'welfare', 'work', 'shirking', 'responsibility', 'seeking', 'responsibility', 'lives', 'let', 'us', 'ask', 'government', 'challenges', 'face', 'together', 'let', 'us', 'ask', 'government', 'help', 'help', 'national', 'government', 'great', 'vital', 'role', 'play', 'pledge', 'government', 'act', 'act', 'boldly', 'lead', 'boldly', 'important', 'role', 'every', 'one', 'us', 'must', 'play', 'individual', 'member', 'community', 'day', 'forward', 'let', 'us', 'make', 'solemn', 'commitment', 'heart', 'bear', 'responsibility', 'part', 'live', 'ideals', 'together', 'see', 'dawn', 'new', 'age', 'progress', 'america', 'together', 'celebrate', '200th', 'anniversary', 'nation', 'proud', 'fulfillment', 'promise', 'world', 'america', 's', 'longest', 'difficult', 'war', 'comes', 'end', 'let', 'us', 'learn', 'debate', 'differences', 'civility', 'decency', 'let', 'us', 'reach', 'one', 'precious', 'quality', 'government', 'provide', 'new', 'level', 'respect', 'rights', 'feelings', 'one', 'another', 'new', 'level', 'respect', 'individual', 'human', 'dignity', 'cherished', 'birthright', 'every', 'american', 'else', 'time', 'come', 'us', 'renew', 'faith', 'america', 'recent', 'years', 'faith', 'challenged', 'children', 'taught', 'ashamed', 'country', 'ashamed', 'parents', 'ashamed', 'america', 's', 'record', 'home', 'role', 'world', 'every', 'turn', 'beset', 'find', 'everything', 'wrong', 'america', 'little', 'right', 'confidence', 'judgment', 'history', 'remarkable', 'times', 'privileged', 'live', 'america', 's', 'record', 'century', 'unparalleled', 'world', 's', 'history', 'responsibility', 'generosity', 'creativity', 'progress', 'let', 'us', 'proud', 'system', 'produced', 'provided', 'freedom', 'abundance', 'widely', 'shared', 'system', 'history', 'world', 'let', 'us', 'proud', 'four', 'wars', 'engaged', 'century', 'including', 'one', 'bringing', 'end', 'fought', 'selfish', 'advantage', 'help', 'others', 'resist', 'aggression', 'let', 'us', 'proud', 'bold', 'new', 'initiatives', 'steadfastness', 'peace', 'honor', 'made', 'break-through', 'toward', 'creating', 'world', 'world', 'known', 'structure', 'peace', 'last', 'merely', 'time', 'generations', 'come', 'embarking', 'today', 'era', 'presents', 'challenges', 'great', 'nation', 'generation', 'ever', 'faced', 'shall', 'answer', 'god', 'history', 'conscience', 'way', 'use', 'years', 'stand', 'place', 'hallowed', 'history', 'think', 'others', 'stood', 'think', 'dreams', 'america', 'think', 'recognized', 'needed', 'help', 'far', 'beyond', 'order', 'make', 'dreams', 'come', 'true', 'today', 'ask', 'prayers', 'years', 'ahead', 'may', 'god', 's', 'help', 'making', 'decisions', 'right', 'america', 'pray', 'help', 'together', 'may', 'worthy', 'challenge', 'let', 'us', 'pledge

e', 'together', 'make', 'next', 'four', 'years', 'best', 'four', 'years', 'america',
 "'s", 'history', '200th', 'birthday', 'america', 'young', 'vital', 'began', 'bright', 'b
 eacon', 'hope', 'world', 'let', 'us', 'go', 'forward', 'confident', 'hope', 'strong', 'f
 aith', 'one', 'another', 'sustained', 'faith', 'god', 'created', 'us', 'striving', 'alwa
 ys', 'serve', 'purpose']

 ['mr.', 'vice', 'president', ',', 'mr.', 'speaker', ',', 'mr.', 'chief', 'justice', ',',
 'senator', 'cook', ',', 'mrs.', 'eisenhower', ',', 'and', 'my', 'fellow', 'citizens', 'o
 f', 'this', 'great', 'and', 'good', 'country', 'we', 'share', 'together', 'when', 'we',
 'met', 'here', 'four', 'years', 'ago', ',', 'america', 'was', 'bleak', 'in', 'spirit',
 ',', 'depressed', 'by', 'the', 'prospect', 'of', 'seemingly', 'endless', 'war', 'abroa
 d', 'and', 'of', 'destructive', 'conflict', 'at', 'home', '.,', 'as', 'we', 'meet', 'her
 e', 'today', ',', 'we', 'stand', 'on', 'the', 'threshold', 'of', 'a', 'new', 'era', 'o
 f', 'peace', 'in', 'the', 'world', '.,', 'the', 'central', 'question', 'before', 'us', 'i
 s', 'how', 'shall', 'we', 'use', 'that', 'peace', '?', 'let', 'us', 'resolve', 'that',
 'this', 'era', 'we', 'are', 'about', 'to', 'enter', 'will', 'not', 'be', 'what', 'othe
 r', 'postwar', 'periods', 'have', 'so', 'often', 'been', 'a', 'time', 'of', 'retreat',
 'and', 'isolation', 'that', 'leads', 'to', 'stagnation', 'at', 'home', 'and', 'invites',
 'new', 'danger', 'abroad', '.,', 'let', 'us', 'resolve', 'that', 'this', 'will', 'be', 'w
 hat', 'it', 'can', 'become', 'a', 'time', 'of', 'great', 'responsibilities', 'greatly',
 'borne', ',', 'in', 'which', 'we', 'renew', 'the', 'spirit', 'and', 'the', 'promise', 'o
 f', 'america', 'as', 'we', 'enter', 'our', 'third', 'century', 'as', 'a', 'nation', '.,',
 'this', 'past', 'year', 'saw', 'far-reaching', 'results', 'from', 'our', 'new', 'policie
 s', 'for', 'peace', '.,', 'by', 'continuing', 'to', 'revitalize', 'our', 'traditional',
 'friendships', ',', 'and', 'by', 'our', 'missions', 'to', 'peking', 'and', 'to', 'mosco
 w', ',', 'we', 'were', 'able', 'to', 'establish', 'the', 'base', 'for', 'a', 'new', 'an
 d', 'more', 'durable', 'pattern', 'of', 'relationships', 'among', 'the', 'nations', 'o
 f', 'the', 'world', '.,', 'because', 'of', 'america', "'s", 'bold', 'initiatives', ',',
 '1972', 'will', 'be', 'long', 'remembered', 'as', 'the', 'year', 'of', 'the', 'greatest
 progress', 'since', 'the', 'end', 'of', 'world', 'war', 'ii', 'toward', 'a', 'lasti
 ng', 'peace', 'in', 'the', 'world', '.,', 'the', 'peace', 'we', 'seek', 'in', 'the', 'wor
 ld', 'is', 'not', 'the', 'flimsy', 'peace', 'which', 'is', 'merely', 'an', 'interlude',
 'between', 'wars', ',', 'but', 'a', 'peace', 'which', 'can', 'endure', 'for', 'generatio
 ns', 'to', 'come', '.,', 'it', 'is', 'important', 'that', 'we', 'understand', 'both', 'th
 e', 'necessity', 'and', 'the', 'limitations', 'of', 'america', "'s", 'role', 'in', 'main
 taining', 'that', 'peace', '.,', 'unless', 'we', 'in', 'america', 'work', 'to', 'preserv
 e', 'the', 'peace', ',', 'there', 'will', 'be', 'no', 'peace', '.,', 'unless', 'we', 'i
 n', 'america', 'work', 'to', 'preserve', 'freedom', ',', 'there', 'will', 'be', 'no', 'f
 reedom', '.,', 'but', 'let', 'us', 'clearly', 'understand', 'the', 'new', 'nature', 'of',
 'america', "'s", 'role', ',', 'as', 'a', 'result', 'of', 'the', 'new', 'policies', 'we',
 'have', 'adopted', 'over', 'these', 'past', 'four', 'years', '.,', 'we', 'shall', 'respec
 t', 'our', 'treaty', 'commitments', '.,', 'we', 'shall', 'support', 'vigorously', 'the',
 'principle', 'that', 'no', 'country', 'has', 'the', 'right', 'to', 'impose', 'its', 'wil
 l', 'or', 'rule', 'on', 'another', 'by', 'force', '.,', 'we', 'shall', 'continue', ',',
 'in', 'this', 'era', 'of', 'negotiation', ',', 'to', 'work', 'for', 'the', 'limitation',
 'of', 'nuclear', 'arms', ',', 'and', 'to', 'reduce', 'the', 'danger', 'of', 'confrontati
 on', 'between', 'the', 'great', 'powers', '.,', 'we', 'shall', 'do', 'our', 'share', 'i
 n', 'defending', 'peace', 'and', 'freedom', 'in', 'the', 'world', '.,', 'but', 'we', 'sha
 ll', 'expect', 'others', 'to', 'do', 'their', 'share', '.,', 'the', 'time', 'has', 'passe
 d', 'when', 'america', 'will', 'make', 'every', 'other', 'nation', "'s", 'conflict', 'ou
 r', 'own', ',', 'or', 'make', 'every', 'other', 'nation', "'s", 'future', 'our', 'respon
 sibility', ',', 'or', 'presume', 'to', 'tell', 'the', 'people', 'of', 'other', 'nation
 s', 'how', 'to', 'manage', 'their', 'own', 'affairs', '.,', 'just', 'as', 'we', 'respec
 t', 'the', 'right', 'of', 'each', 'nation', 'to', 'determine', 'its', 'own', 'future',
 ',', 'we', 'also', 'recognize', 'the', 'responsibility', 'of', 'each', 'nation', 'to',
 'secure', 'its', 'own', 'future', '.,', 'just', 'as', 'america', "'s", 'role', 'is', 'ind
 ispensable', 'in', 'preserving', 'the', 'world', "'s", 'peace', ',', 'so', 'is', 'each',
 'nation', "'s", 'role', 'indispensable', 'in', 'preserving', 'its', 'own', 'peace', '.,',
 'together', 'with', 'the', 'rest', 'of', 'the', 'world', ',', 'let', 'us', 'resolve', 't
 o', 'move', 'forward', 'from', 'the', 'beginnings', 'we', 'have', 'made', '.,', 'let', 'u
 s', 'continue', 'to', 'bring', 'down', 'the', 'walls', 'of', 'hostility', 'which', 'hav

e', 'divided', 'the', 'world', 'for', 'too', 'long', ',', 'and', 'to', 'build', 'in', 'the', 'place', 'bridges', 'of', 'understanding', 'so', 'that', 'despite', 'profound', 'differences', 'between', 'systems', 'of', 'government', ',', 'the', 'people', 'of', 'the', 'world', 'can', 'be', 'friends', '.,', 'let', 'us', 'build', 'a', 'structure', 'of', 'peace', 'in', 'the', 'world', 'in', 'which', 'the', 'weak', 'are', 'as', 'safe', 'as', 'the', 'strong', 'in', 'which', 'each', 'respects', 'the', 'right', 'of', 'the', 'other', 'to', 'live', 'by', 'a', 'different', 'system', 'in', 'which', 'those', 'who', 'would', 'influence', 'others', 'will', 'do', 'so', 'by', 'the', 'strength', 'of', 'their', 'ideas', ',', 'and', 'not', 'by', 'the', 'force', 'of', 'their', 'arms', '.,', 'let', 'us', 'accept', 'that', 'high', 'responsibility', 'not', 'as', 'a', 'burden', ',', 'but', 'gladly', 'gladly', 'because', 'the', 'chance', 'to', 'build', 'such', 'a', 'peace', 'is', 'the', 'noblest', 'endeavor', 'in', 'which', 'a', 'nation', 'can', 'engage', 'gladly', '.,', 'also', '.,', 'because', 'only', 'if', 'we', 'act', 'greatly', 'in', 'meeting', 'our', 'responsibilities', 'abroad', 'will', 'we', 'remain', 'a', 'great', 'nation', '.,', 'and', 'only', 'if', 'we', 'remain', 'a', 'great', 'nation', 'will', 'we', 'act', 'greatly', 'in', 'meeting', 'our', 'challenges', 'at', 'home', '.,', 'we', 'have', 'the', 'chance', 'today', 'to', 'do', 'more', 'than', 'ever', 'before', 'in', 'our', 'history', 'to', 'make', 'life', 'better', 'in', 'america', 'to', 'ensure', 'better', 'education', '.,', 'better', 'health', '.,', 'better', 'housing', '.,', 'better', 'transportation', '.,', 'a', 'cleaner', 'environment', 'to', 'restore', 'respect', 'for', 'law', '.,', 'to', 'make', 'our', 'communities', 'more', 'livable', 'and', 'to', 'insure', 'the', 'god-given', 'right', 'of', 'every', 'american', 'to', 'full', 'and', 'equal', 'opportunity', '.,', 'because', 'the', 'range', 'of', 'our', 'needs', 'is', 'so', 'great', 'because', 'the', 'reach', 'of', 'our', 'opportunities', 'is', 'so', 'great', 'let', 'us', 'be', 'bold', 'in', 'our', 'determination', 'to', 'meet', 'those', 'needs', 'in', 'new', 'ways', '.,', 'just', 'as', 'building', 'a', 'structure', 'of', 'peace', 'abroad', 'has', 'required', 'turning', 'away', 'from', 'old', 'policies', 'that', 'failed', '.,', 'so', 'building', 'a', 'new', 'era', 'of', 'progress', 'at', 'home', 'requires', 'turning', 'away', 'from', 'old', 'policies', 'that', 'have', 'failed', '.,', 'abroad', '.,', 'the', 'shift', 'from', 'old', 'policies', 'to', 'new', 'has', 'not', 'been', 'a', 'retreat', 'from', 'our', 'responsibilities', '.,', 'but', 'a', 'better', 'way', 'to', 'peace', '.,', 'and', 'at', 'home', '.,', 'the', 'shift', 'from', 'old', 'policies', 'to', 'new', 'will', 'not', 'be', 'a', 'retreat', 'from', 'our', 'responsibilities', '.,', 'but', 'a', 'better', 'way', 'to', 'progress', '.,', 'abroad', 'and', 'at', 'home', '.,', 'the', 'key', 'to', 'those', 'new', 'responsibilities', 'lies', 'in', 'the', 'placing', 'and', 'the', 'division', 'of', 'responsibility', '.,', 'we', 'have', 'lived', 'too', 'long', 'with', 'the', 'consequences', 'of', 'attempting', 'to', 'gather', 'all', 'power', 'and', 'responsibility', 'in', 'washington', '.,', 'abroad', 'and', 'at', 'home', '.,', 'the', 'time', 'has', 'come', 'to', 'turn', 'away', 'from', 'the', 'condescending', 'policies', 'of', 'paternalism', 'of', '.,', 'washington', 'knows', 'best', '.,', 'a', 'person', 'can', 'be', 'expected', 'to', 'act', 'responsibly', 'only', 'if', 'he', 'has', 'responsibility', '.,', 'this', 'is', 'human', 'nature', '.,', 'so', 'let', 'us', 'encourage', 'individuals', 'at', 'home', 'and', 'nations', 'abroad', 'to', 'do', 'more', 'for', 'themselves', '.,', 'to', 'decide', 'more', 'for', 'themselves', '.,', 'let', 'us', 'locate', 'responsibility', 'in', 'more', 'places', '.,', 'let', 'us', 'measure', 'what', 'we', 'will', 'do', 'for', 'others', 'by', 'what', 'they', 'will', 'do', 'for', 'themselves', '.,', 'that', 'is', 'why', 'today', 'i', 'offer', 'no', 'promise', 'of', 'a', 'purely', 'governmental', 'solution', 'for', 'every', 'problem', '.,', 'we', 'have', 'lived', 'too', 'long', 'with', 'that', 'false', 'promise', '.,', 'in', 'trusting', 'too', 'much', 'in', 'government', '.,', 'we', 'have', 'asked', 'of', 'it', 'more', 'than', 'it', 'can', 'deliver', '.,', 'this', 'leads', 'only', 'to', 'inflated', 'expectations', '.,', 'to', 'reduced', 'individual', 'effort', '.,', 'and', 'to', 'a', 'disappointment', 'and', 'frustration', 'that', 'erode', 'confidence', 'both', 'in', 'what', 'government', 'can', 'do', 'and', 'in', 'what', 'people', 'can', 'do', '.,', 'government', 'must', 'learn', 'to', 'take', 'less', 'from', 'people', 'so', 'that', 'people', 'can', 'do', 'more', 'for', 'themselves', '.,', 'let', 'us', 'remember', 'that', 'america', 'was', 'built', 'not', 'by', 'government', '.,', 'but', 'by', 'people', 'not', 'by', 'welfare', '.,', 'but', 'by', 'work', 'not', 'by', 'shirking', 'responsibility', '.,', 'but', 'by', 'seeking', 'responsibility', '.,', 'in', 'our', 'own', 'lives', '.,', 'let', 'each', 'of', 'us', 'ask', 'not', 'just', 'what', 'will', 'government', 'do', 'for', 'me', '.,', 'but', 'what', 'can', 'i', 'do', 'for', 'myself', '.,', 'in', 'the', 'challenges', 'we', 'face', 'together', '.,', 'let', 'each', 'of',

'us', 'ask', 'not', 'just', 'how', 'can', 'government', 'help', 'but', 'how', 'can', 'i', 'help', 'your', 'national', 'government', 'has', 'a', 'great', 'and', 'vital', 'role', 'to', 'play', 'and', 'i', 'pledge', 'to', 'you', 'that', 'where', 'this', 'government', 'should', 'act', 'we', 'will', 'act', 'boldly', 'and', 'we', 'will', 'lead', 'boldly', 'but', 'just', 'as', 'important', 'is', 'the', 'role', 'that', 'each', 'and', 'every', 'one', 'of', 'us', 'must', 'play', 'as', 'an', 'individual', 'and', 'as', 'a', 'member', 'of', 'his', 'own', 'community', 'from', 'this', 'day', 'forward', 'let', 'each', 'of', 'us', 'make', 'a', 'solemn', 'commitment', 'in', 'his', 'own', 'heart', 'to', 'bear', 'his', 'responsibility', 'to', 'do', 'his', 'part', 'to', 'live', 'his', 'ideals', 'so', 'that', 'together', 'we', 'can', 'see', 'the', 'dawn', 'of', 'a', 'new', 'age', 'of', 'progress', 'for', 'america', 'and', 'together', 'as', 'we', 'celebrate', 'our', '200th', 'anniversary', 'as', 'a', 'nation', 'we', 'can', 'do', 'so', 'proud', 'in', 'the', 'fulfillment', 'of', 'our', 'promise', 'to', 'ourselves', 'and', 'to', 'the', 'world', 'as', 'america', 'is', 'the', 'longest', 'and', 'most', 'difficult', 'war', 'comes', 'to', 'an', 'end', 'let', 'us', 'again', 'learn', 'to', 'debate', 'our', 'differences', 'with', 'civility', 'and', 'decency', 'and', 'let', 'each', 'of', 'us', 'reach', 'out', 'for', 'that', 'one', 'precious', 'quality', 'government', 'can', 'not', 'provide', 'a', 'new', 'level', 'of', 'respect', 'for', 'the', 'rights', 'and', 'feelings', 'of', 'one', 'another', 'a', 'new', 'level', 'of', 'respect', 'for', 'the', 'individual', 'human', 'dignity', 'which', 'is', 'the', 'cherished', 'birthright', 'of', 'every', 'american', 'above', 'all', 'else', 'the', 'time', 'has', 'come', 'for', 'us', 'to', 'renew', 'our', 'faith', 'in', 'ourselves', 'and', 'in', 'america', 'in', 'recent', 'years', 'that', 'faith', 'has', 'been', 'challenged', 'our', 'children', 'have', 'been', 'taught', 'to', 'be', 'ashamed', 'of', 'their', 'country', 'ashamed', 'of', 'their', 'parents', 'ashamed', 'of', 'america', 's', 'record', 'at', 'home', 'and', 'of', 'its', 'role', 'in', 'the', 'world', 'at', 'every', 'turn', 'we', 'have', 'been', 'beset', 'by', 'those', 'who', 'find', 'everything', 'wrong', 'with', 'america', 'and', 'little', 'that', 'is', 'right', 'but', 'i', 'am', 'confident', 'that', 'this', 'will', 'not', 'be', 'the', 'judgment', 'of', 'history', 'on', 'these', 'remarkable', 'times', 'in', 'which', 'we', 'are', 'privileged', 'to', 'live', 'america', 's', 'record', 'in', 'this', 'century', 'has', 'been', 'unparalleled', 'in', 'the', 'world', 's', 'history', 'for', 'its', 'responsibility', 'for', 'its', 'generosity', 'for', 'its', 'creativity', 'and', 'for', 'its', 'progress', 'let', 'us', 'be', 'proud', 'that', 'our', 'system', 'has', 'produced', 'and', 'provided', 'more', 'freedom', 'and', 'more', 'abundance', 'more', 'widely', 'shared', 'than', 'any', 'other', 'system', 'in', 'the', 'history', 'of', 'the', 'world', 'let', 'us', 'be', 'proud', 'that', 'in', 'each', 'of', 'the', 'four', 'wars', 'in', 'which', 'we', 'have', 'been', 'engaged', 'in', 'this', 'century', 'including', 'the', 'one', 'we', 'are', 'now', 'bringing', 'to', 'an', 'end', 'we', 'have', 'fought', 'not', 'for', 'our', 'selfish', 'advantage', 'but', 'to', 'help', 'others', 'resist', 'aggression', 'let', 'us', 'be', 'proud', 'that', 'by', 'our', 'bold', 'new', 'initiatives', 'and', 'by', 'our', 'steadfastness', 'for', 'peace', 'with', 'honor', 'we', 'have', 'made', 'a', 'break-through', 'toward', 'creating', 'in', 'the', 'world', 'what', 'the', 'world', 'has', 'not', 'known', 'before', 'a', 'structure', 'of', 'peace', 'that', 'can', 'last', 'not', 'merely', 'for', 'our', 'time', 'but', 'for', 'generations', 'to', 'come', 'we', 'are', 'embarking', 'here', 'today', 'on', 'an', 'era', 'that', 'presents', 'challenges', 'great', 'as', 'those', 'any', 'nation', 'or', 'any', 'generation', 'has', 'ever', 'faced', 'we', 'shall', 'answer', 'to', 'god', 'to', 'history', 'and', 'to', 'our', 'conscience', 'for', 'the', 'way', 'in', 'which', 'we', 'use', 'these', 'years', 'as', 'i', 'stand', 'in', 'this', 'place', 'so', 'hallowed', 'by', 'history', 'i', 'think', 'of', 'others', 'who', 'have', 'stood', 'here', 'before', 'me', 'i', 'think', 'of', 'the', 'dreams', 'they', 'had', 'for', 'america', 'and', 'i', 'think', 'of', 'how', 'each', 'recognized', 'that', 'he', 'needed', 'help', 'far', 'beyond', 'himself', 'in', 'order', 'to', 'make', 'those', 'dreams', 'come', 'true', 'today', 'i', 'ask', 'your', 'prayers', 'that', 'in', 'the', 'years', 'ahead', 'i', 'may', 'have', 'god', 's', 'help', 'in', 'making', 'decisions', 'that', 'are', 'right', 'for', 'america', 'and', 'i', 'pray', 'for', 'your', 'help', 'so', 'that', 'together', 'we', 'may', 'be', 'worthy', 'of', 'our', 'challenge', 'let', 'us', 'pledge', 'together', 'to', 'make', 'these', 'next', 'four', 'years', 'the', 'best', 'four', 'years', 'in', 'america', 's', 'history', 's

```
o', 'that', 'on', 'its', '200th', 'birthday', 'america', 'will', 'be', 'as', 'young', 'a
nd', 'as', 'vital', 'as', 'when', 'it', 'began', ',', 'and', 'as', 'bright', 'a', 'beaco
n', 'of', 'hope', 'for', 'all', 'the', 'world', '.', 'let', 'us', 'go', 'forward', 'fro
m', 'here', 'confident', 'in', 'hope', ',', 'strong', 'in', 'our', 'faith', 'in', 'one',
'another', ',', 'sustained', 'by', 'our', 'faith', 'in', 'god', 'who', 'created', 'us',
',', 'and', 'striving', 'always', 'to', 'serve', 'his', 'purpose', '.']
```

```
In [131... print(len(Nixon_without_stopwords_punc))

print(len(N_tokens))
```

```
848
1983
```

2.3 Which word occurs the most number of times in his inaugural address for each president? Mention the top three words. (after removing the stopwords)

```
In [132... #Checking most occuring words

word_freq_Roosevelt = nltk.FreqDist(Roosevelt_without_stopwords_punc)

word_freq_Roosevelt.most_common(3)
```

```
Out[132... [('nation', 12), ('know', 10), ('spirit', 9)]
```

```
In [134... word_freq_Kennedy = nltk.FreqDist(Kennedy_without_stopwords_punc)

word_freq_Kennedy.most_common(3)
```

```
Out[134... [('let', 16), ('us', 12), ('world', 8)]
```

```
In [135... word_freq_Nixon = nltk.FreqDist(Nixon_without_stopwords_punc)

word_freq_Nixon.most_common(3)
```

```
Out[135... [('us', 26), ('let', 22), ('america', 21)]
```

2.4 Plot the word cloud of each of the speeches of the variable. (after removing the stopwords)

```
In [133... # Import package
import matplotlib.pyplot as plt

# Defining a function to plot word cloud
def plot_cloud(wordcloud):
    plt.figure(figsize=(40, 30))
    plt.imshow(wordcloud)
    plt.axis("off");
```

```
In [138... # Import package
from wordcloud import WordCloud, STOPWORDS
```

In [150...

```

Ro_string = str(Roosevelt_without_stopwords_punc)
Ke_string = str(Kennedy_without_stopwords_punc)
Ni_string = str(Nixon_without_stopwords_punc)

```

```

#removing punctuations

```

```

re.sub("[^9A-Za-z ]", "", Ro_string)
re.sub("[^9A-Za-z ]", "", Ke_string)
re.sub("[^9A-Za-z ]", "", Ni_string)

```

Out[150...

```

'mr vice president mr speaker mr chief justice senator cook mrs eisenhower fellow citize
ns great good country share together met four years ago america bleak spirit depressed p
rospect seemingly endless war abroad destructive conflict home meet today stand threshol
d new era peace world central question us shall use peace let us resolve era enter postw
ar periods often time retreat isolation leads stagnation home invites new danger abroad
let us resolve become time great responsibilities greatly borne renew spirit promise ame
rica enter third century nation past year saw far-reaching results new policies peace co
ntinuing revitalize traditional friendships missions peking moscow able establish base n
ew durable pattern relationships among nations world america s bold initiatives 9 long r
emembered year greatest progress since end world war ii toward lasting peace world peace
seek world flimsy peace merely interlude wars peace endure generations come important un
derstand necessity limitations america s role maintaining peace unless america work pres
erve peace peace unless america work preserve freedom freedom let us clearly understand
new nature america s role result new policies adopted past four years shall respect trea
ty commitments shall support vigorously principle country right impose rule another forc
e shall continue era negotiation work limitation nuclear arms reduce danger confrontatio
n great powers shall share defending peace freedom world shall expect others share time
passed america make every nation s conflict make every nation s future responsibility pr
esume tell people nations manage affairs respect right nation determine future also reco
gnize responsibility nation secure future america s role indispensable preserving world
s peace nation s role indispensable preserving peace together rest world let us resolve
move forward beginnings made let us continue bring walls hostility divided world long bu
ild place bridges understanding despite profound differences systems government people w
orld friends let us build structure peace world weak safe strong respects right live dif
ferent system would influence others strength ideas force arms let us accept high respon
sibility burden gladly gladly chance build peace noblest endeavor nation engage gladly a
lso act greatly meeting responsibilities abroad remain great nation remain great nation
act greatly meeting challenges home chance today ever history make life better america e
nsure better education better health better housing better transportation cleaner enviro
nment restore respect law make communities livable insure god-given right every american
full equal opportunity range needs great reach opportunities great let us bold determina
tion meet needs new ways building structure peace abroad required turning away old polic
ies failed building new era progress home requires turning away old policies failed abro
ad shift old policies new retreat responsibilities better way peace home shift old polic
ies new retreat responsibilities better way progress abroad home key new responsibilitie
s lies placing division responsibility lived long consequences attempting gather power r
esponsibility washington abroad home time come turn away condescending policies paternal
ism washington knows best person expected act responsibly responsibility human nature
let us encourage individuals home nations abroad decide let us locate responsibility pla
ces let us measure others today offer promise purely governmental solution every problem
lived long false promise trusting much government asked deliver leads inflated expectati
ons reduced individual effort disappointment frustration erode confidence government peo
ple government must learn take less people people let us remember america built governme
nt people welfare work shirking responsibility seeking responsibility lives let us ask g
overnment challenges face together let us ask government help help national government g
reat vital role play pledge government act act boldly lead boldly important role every o
ne us must play individual member community day forward let us make solemn commitment he
art bear responsibility part live ideals together see dawn new age progress america toge

```



```
wordcloud = WordCloud(width = 2500, height = 2000, random_state=1, background_color='sa
# Plot
plot_cloud(wordcloud)
```



THE END

In []: